

An AI-Ready Air-to-Ground Channel Model for HAPS–UAV Access Networks

Channel-Model Implementation and Signal-Quality Characterisation

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Abstract—This paper presents a simulation and validation study for an AI-driven HAPS–UAV–Ground access network, addressing challenges like SINR heterogeneity induced by a fixed 3-HAPS equilateral-triangle constellation across its service footprint. Following Arani et al., we implement (i) a free-space model for the High-Altitude Platform Station (HAPS) link, (ii) building-blockage line-of-sight (LoS) probability series for the Unmanned-Aerial-Vehicle (UAV) links, (iii) average path loss and downlink signal-to-interference-plus-noise ratio (SINR) under this constellation geometry, and (iv) comprehensive ITU-R atmospheric losses (rain, fog, gas, scintillation) for backhaul and feeder-link design. The channel model is realised in Python (NumPy + Matplotlib + itur package) and characterised through signal-quality curves, spatial-heterogeneity analysis, and atmospheric sensitivity results. An LSTM forecasts 38 GHz feeder rain attenuation up to six hours ahead; on held-out 2026 data it is more accurate than persistence only at four to six hours (RMSE), and the DQN uses its one-hour-ahead forecast. A DQN agent using this forecast earns higher reward and lower outage than max-SINR across five seeds, but lower fairness.

Index Terms—HAPS, UAV, non-terrestrial networks, air-to-ground channel, LoS probability, SINR, multi-node networks, UAV relay, backhaul link, feeder link, beam tracking, rain attenuation, system geometry

I. INTRODUCTION

A. Platform Overview

A typical HAPS system consists of a stratospheric, solar-powered platform operating at approximately 20 km altitude. At this altitude, it maintains a fixed service footprint while undergoing continuous circular flight and attitude variations (roll, pitch, and yaw), supporting a service radius of up to 100 km [1], [2]. It communicates with ground equipment through

- **Backhaul link:** HAPS to Gateway
Backhaul link provides high-capacity connectivity between the Ground Gateway (GW) and the HAPS aircraft. Backhaul links operate in the 38.0–39.5 GHz Ka-band 3GPP NTN [12].
- **Service link:** HAPS to User
Service links are designed for connectivity from HAPS directly to UE, ensuring accessibility without specialized hardware for a long range. Service links operate in the 2–2.2 GHz FDD band [12].
- **Feeder link:** HAPS to UAV
Feeder link connects the HAPS aircraft to the relaying UAV, forming the first leg of the relay path below. Feeder links operate in the 38.0–39.5 GHz Ka-band 3GPP NTN [12], the same band as the backhaul link.

- **Relay Link:** HAPS to UAV to User

Relay links have two legs. The first leg (the feeder link above) connects the HAPS to the UAV, and the second leg connects the UAV to the ground UEs with the UAV positioned at a lower altitude. Relay links serve as the primary backup path for users when direct HAPS-to-UE service links are unavailable. Relay links operate in 38.0–39.5 GHz Ka-band 3GPP NTN [12] for the HAPS-UAV feeder leg and the 2–2.2 GHz FDD band [12] for the UAV-User leg.

B. Challenges and Impacts on Connectivity

1) Atmospheric Losses:

- The 38.0–39.5 GHz link (Ka-band) incurs significant atmospheric loss from rain (specific attenuation ≈ 3.0 – 6.8 dB/km at 10–25 mm/h per ITU-R P.838 [5]), fog [7], and gas absorption [6]. The 38 GHz band is still chosen as it offers practical spectrum allocation, avoiding the 22.235 GHz water-vapor absorption resonance [6] and the 60 GHz oxygen peak [6].
- The 2 GHz link exhibits high resilience to atmospheric effects according to ITU-R P.618 [22], providing a stable “last-mile” connection even under adverse weather. But it is sensitive to building blockage, which can cause significant path loss.

2) Elevation Angle and Slant Path:

Because the HAPS provides coverage up to a radius of 100 km [2], many links operate at low elevation angles. These “slant paths” (§2.1, worst-case 102 km) result in a significantly longer signal trajectory through the moisture-dense troposphere compared to vertical (nadir) paths, leading to higher cumulative attenuation per ITU-R P.618 [22].

C. Problem Statement

The above considerations define the control-layer problem for relay links: determining the position of each UAV relay and assigning service or relay channels so that users are served fairly and reliably. The system must also select the best serving HAPS for either the HAPS-to-UAV or HAPS-to-UE path to maximize throughput and minimize outage. The control layer must adapt to changing user positions, weather conditions, and HAPS battery states.

In this document, we (i) characterise the channel: free-space and LoS-probability path loss, SINR under the 3-

TABLE I: POC Network Configuration (AGL = Above Ground Level)

Node	Altitude	Count	Spacing
HAPS	20 km	3	Equilateral 65km, 1 Server, 2 Interferer
UAV Relay	500 m AGL	1	Static (no mobility)
Gateway	50 m AGL	1	Centroid of HAPS triangle
Users	1.5 m (hand-held)	N (configurable)	Uniform random in coverage

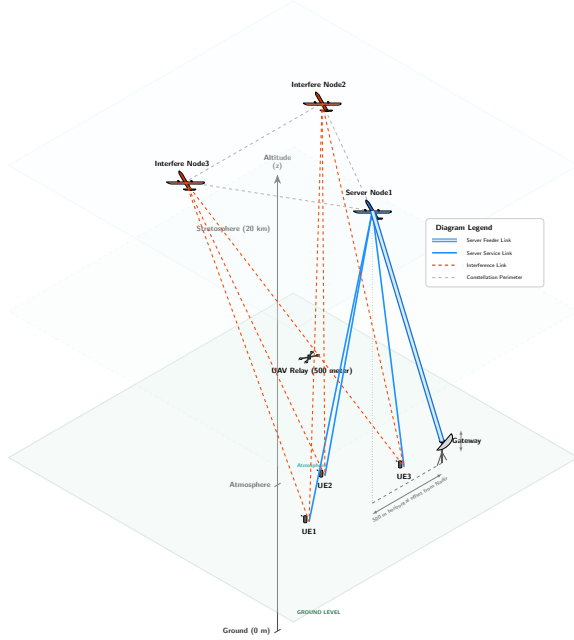


Fig. 1: Topology

HAPS geometry, and ITU-R atmospheric losses at 38 GHz; (ii) compare two association rules, random and max-SINR, which select the platform for each user; (iii) build an LSTM that forecasts the 38 GHz feeder rain attenuation over the next six hours from past weather; and (iv) train a DQN agent that uses this forecast in its state, comparing it with max-SINR and random on 50 held-out days from 2025 over five seeds. UAV placement is analysed through relay geometry and the landscape LoS mechanism; it is not optimised.

II. POC - TOPOLOGY AND LINK TYPES

A. Network Topology

- 1) **Access layer:** 3 HAPS nodes + 1 UAV relay serve ground users.
- 2) **Backhaul layer:** 1 ground gateway aggregates backhaul links from all 3 HAPS.
- 3) **Core network:** Gateway connects to Internet.

B. Backhaul Link Geometry

Backhaul links operate in the 38.0–39.5 GHz millimeter-wave band allocated to HAPS by ITU-R M.2101 (Ka-band) [1]. The backhaul link is a short-range connection from the HAPS aircraft to the ground-based Gateway.

- Gateway Altitude: 50 m AGL
- Altitude difference: $20 \text{ km} - 0.05 \text{ km} = 19.95 \text{ km}$
- Horizontal distance: 50 m from HAPS nadir.
- Elevation angle: $\theta = \arctan\left(\frac{19.95}{0.05}\right) \approx 89.86^\circ$ (near zenith)
- Slant path: $L_{\text{slant}} = \sqrt{19.95^2 + 0.05^2} \approx 19.95 \text{ km}$
- HAPS transmit power: $P_h = 43 \text{ dBm}$; HAPS-side transmit antenna gain: $G_h = 13.8 \text{ dBi}$ (directional Ka-band aperture toward the Gateway, per the 30 GHz-class HAPS link budget of Karaman et al. [14])
- Gateway receive antenna gain: $G_g = 39.7 \text{ dBi}$ (fixed VSAT-class terminal, larger aperture feasible at a ground installation [14])
- Combined link EIRP + receive gain: $P_h + G_h + G_g \approx 96.5 \text{ dBm}$ -equivalent, before FSPL and atmospheric losses (§III.B)

C. Relay Link Geometry

The relay link (HAPS-to-UAV 38.0–39.5 GHz and UAV-to-User, $f = 2.1 - 2.2 \text{ GHz}$ [1]) connects HAPS to ground users via the UAV relay node positioned at 500 m AGL (Table I).

1) HAPS-to-UAV leg (Feeder Link, 38 GHz):

- UAV Relay Altitude: 500 m AGL
- Altitude difference: $20 \text{ km} - 0.5 \text{ km} = 19.5 \text{ km}$
- Horizontal distance: 500 m from HAPS nadir.
- Elevation angle: $\theta = \arctan\left(\frac{19.5}{0.5}\right) \approx 88.5^\circ$ (near zenith)
- Slant path: $L_{\text{slant}} = \sqrt{19.5^2 + 0.5^2} \approx 19.51 \text{ km}$
- HAPS transmit power: $P_h = 43 \text{ dBm}$; HAPS-side transmit antenna gain: $G_h = 13.8 \text{ dBi}$ (same directional Ka-band aperture as the backhaul link, §II-B)
- UAV receive antenna gain: $G_u = 30.7 \text{ dBi}$ (UAV-mountable 32×32 -element mmWave patch array, conservative low end of the 30.7–32.8 dBi range reported at 25.4–26.9 GHz [15], applied here at 38 GHz)

2) UAV-to-User leg (2.1 – 2.2 GHz):

- User Altitude: 1.5 m (hand-held)
- Altitude difference: $0.5 \text{ km} - 0.0015 \text{ km} = 0.4985 \text{ km}$
- Horizontal distance: 500 m to 100 km from UAV.
- Elevation angle: For 500m horizontal distance from UAV, $\theta = \arctan\left(\frac{0.4985}{0.5}\right) \approx 44.9^\circ$, for larger distances, the angle becomes much smaller.
- Slant path: $L_{\text{slant}} = \sqrt{0.4985^2 + 0.5^2} \approx 0.707 \text{ km}$
- UAV transmit power: $P_u = 24 \text{ dBm}$; no directional antenna gain is modeled on this leg (UAV-side and UE-side antennas are treated as isotropic, consistent with the service link).

The second leg of relay link is dominated by horizontal distance and building blockage rather than altitude. The attenuation grows sharply once the horizontal offset exceeds the LoS-to-NLoS transition set by local building density and built-up ratio (§III.C, Eq. (8)–(9)). This is why the UAV relay's placement – altitude and horizontal offset from served users – depends on local landscape/building statistics. .

D. Service Link Geometry

Service links operate in the 2.1 – 2.2 GHz FDD band (S-band) [12]. Service link connects ground users directly to a HAPS serving node at 20 km altitude via a slant path extending up to the ~ 100 km service radius. Although the service link shares the 2 GHz band with the relay link, its power budget must close over a much longer range with a mostly-unobstructed, near-zenith-dominated geometry.

- HAPS transmit power: $P_h = 43$ dBm (same platform transmitter as the backhaul and feeder links; P_TX_HAPS_DBM).
- No directional antenna gain is modeled on the service link (HAPS-side and UE-side antennas are treated as isotropic); unlike the 38 GHz backhaul/feeder links, the service link's wide-area, long-range coverage requirement is not compatible with a narrow, high-gain beam.

III. SYSTEM AND CHANNEL MODEL

This section presents three channel models for HAPS to Ground Scenario: (A) generic propagation formulae applicable to all frequency bands, (B) the 38 GHz backhaul link model (also reused for the 38 GHz feeder link, HAPS-to-UAV, in §IV-E), and (C) the 2 GHz service/relay link model and finally

A. Common System Models (All Frequency Bands)

1) *Equation (1) — Free-Space Path Loss [3]*: FSPL is calculated on the basis of a 20 km vertical zenith distance and a lateral reach extending to 100 km. The ITU-R P.530 [3] free-space path loss formula FSPL(MHz, km) is defined as

$$L_{\text{FSPL}} = 32.44 + 20 \log_{10} f_{\text{MHz}} + 20 \log_{10} d_{\text{km}} \text{ [dB]}. \quad (1)$$

f_{MHz} is the frequency of the HAPS carrier in MHz and d_{km} is the distance in km. This formula applies for all frequencies and distances in the HAPS service footprint, establishing the baseline attenuation before atmospheric and propagation effects. Used in the backhaul-link (38 GHz, §III.B) and service-link (2 GHz, §III.C) budgets, and reused for the feeder link (HAPS-to-UAV, 38 GHz, §IV-E). This is algebraically the same free-space loss as the wavelength form $L_{\text{FSPL}}(h, g) = 20 \log_{10}(4\pi d_{h,g}/\lambda)$ used later in the multi-HAPS SINR derivation (Eq. (17) et seq., §III.D): substituting $\lambda = c/f$ and converting to MHz/km units folds the resulting constant $20 \log_{10}(4\pi \times 10^9/c)$ into the 32.44 dB term above.

B. 38 GHz Backhaul Link Model (HAPS-to-Gateway)

The backhaul link (HAPS to Ground Gateway, 38–39.5 GHz Ka-band per ITU-R M.2101). This path is vulnerable to atmospheric attenuation from rain, fog, gas absorption, and scintillation. Section III.B models each component sequentially; the same model is reused for the feeder link (HAPS-to-UAV, also 38–39.5 GHz) in §IV-E.

1) *Equation (2) — Specific Rain Attenuation [5]*: Rain attenuation γ_R (dB/km) is calculated from the local rain rate R (mm/h) using the empirical ITU-R P.838 coefficients [5] k, α :

$$\gamma_R = k(f, \theta) R^{\alpha(f, \theta)} \text{ [dB/km]}, \quad (2)$$

For 38 GHz and 11.3° elevation, the horizontal-polarisation coefficients computed with the `itur` implementation of ITU-R P.838 are $k \approx 0.400$ and $\alpha \approx 0.881$, giving $\gamma_R \approx 3.04$ dB/km at 10 mm/h and ≈ 6.82 dB/km at 25 mm/h [5]. These values represent the power-law relationship between rain rate and specific attenuation.

When we multiply the effective path length L_{eff} (km) by this, we obtain the rain loss:

$$A_{\text{rain}} = \gamma_R L_{\text{eff}} \text{ [dB]}, \quad (3)$$

2) *Equation (3) — Total Rain Attenuation [22]*: The effective path length L_{eff} follows ITU-R P.618 [22] (slant-path reduction factors and site-dependent scaling for non-uniform rain along the path). The specific attenuation above is reported directly: for the 38 GHz, 11.3° worst-case backhaul path (102 km slant), $\gamma_R \approx 3.0$ dB/km at 10 mm/h and ≈ 6.8 dB/km at 25 mm/h. Multiplying by L_{eff} gives the path attenuation for a given rain rate. For the gateway geometry (89.86° , 5.2 km of rain), the path attenuation at a given rate is $\gamma_R L_s r$, where L_s is the slant length through the rain layer and r is the P.618 reduction factor. For Delhi, the ITU-R P.837 rain rate exceeded 0.1 % of the time is $R_{0.1} = 15.7$ mm/h, which gives ≈ 19 dB of rain loss on the gateway path. The clear-sky gateway SNR is about 46 dB, so after rain at 0.1 % about 27 dB remains. The gaseous loss on the gateway path is ≈ 0.4 dB (ITU-R P.676, $\rho = 10 \text{ g/m}^3$), leaving about 26 dB. The 38 GHz gateway link therefore closes at 99.9 % availability. The 0.01 % rate ($R_{0.01} = 61.8$ mm/h) would give ≈ 56 dB and is not the design target.

3) *Equation (4) — Fog/Cloud Attenuation [7]*: Fog attenuation depends on the liquid-water density M [g/m^3], frequency f , and temperature T per ITU-R P.840 [7]. It is negligible below about 10 GHz, becomes noticeable in the millimeter-wave bands, and can be several dB in dense fog. Typical values are approximately 0.05 g/m^3 for medium fog and 0.5 g/m^3 for dense fog [7]. Cloud/fog attenuation is modeled as a specific attenuation γ_c (dB/km) times the slant-path length L_{slant} (km):

$$A_{\text{cloud}} = \gamma_c(f, M, T) L_{\text{slant}} \text{ [dB]}, \quad (4)$$

where γ_c depends on the liquid-water density M [g/m^3], frequency and, weakly, on temperature. Typical liquid-water densities are $M \approx 0.05$ for medium fog and $M \approx 0.5$ for dense fog; γ_c is small below 10 GHz and grows into the millimeter-wave bands [7]. For the 38 GHz backhaul link, fog contributes negligibly to total loss (typically < 0.2 dB).

4) *Equation (5) — Atmospheric Gas Absorption [6]*: Atmospheric gas attenuation combines oxygen and water-vapour specific attenuations γ_o and γ_w (dB/km) computed from local pressure p , temperature T and water-vapour density ρ_w according to ITU-R P.676 [6]. The total gas loss is

$$A_{\text{gas}} = (\gamma_o(p, T, f) + \gamma_w(\rho_w, T, f)) L_{\text{slant}} \text{ [dB]}, \quad (5)$$

Combines oxygen and water-vapor components according to ITU-R P.676 Sections 3.1 and 3.2 [6]. The strongest absorption lines occur at 22.235 GHz and 183 GHz for water vapor,

TABLE II: Atmospheric Effects and ITU-R Recommendations for HAPS backhaul/feeder links (38 GHz)

Atmospheric Effect	Affected Frequency	ITU Recommendation
Rain attenuation	>1 GHz	ITU-R P.618 (Earth-space path prediction), P.837 (rain climate statistics / rain rate), P.838 (specific attenuation)
Clouds and fog	1–200 GHz	ITU-R P.840
Atmospheric gases (oxygen and water vapour)	1–1000 GHz	ITU-R P.676
Atmospheric profiles (pressure, temperature, humidity)	all bands	ITU-R P.835
Surface water vapour statistics	all bands	ITU-R P.836
Tropospheric scintillation	1–100 GHz	ITU-R P.618

and near 59.8 GHz and 118.75 GHz for oxygen [6] (P.676 Sect. 3.1–3.2, Figure 2). This makes Ka/Q/V-band HAPS backhaul/feeder bands much more sensitive to moisture than the sub-6 GHz access links. At 38 GHz, clear-sky gas absorption is approximately 0.2–0.3 dB/km, orders of magnitude lower than at the 60 GHz oxygen resonance (~ 15 –20 dB/km), justifying the 38 GHz band choice per ITU-R analysis [6].

5) *Equation (6) — Scintillation Fade Depth [22]*: Tropospheric scintillation refers to rapid fluctuations in signal amplitude and phase caused by small-scale refractive-index irregularities in the lower troposphere per ITU-R P.618 [22]. In Ka-band (around 38 GHz) backhaul/feeder links, scintillation can cause dynamic fading even in clear-sky conditions and must be taken into account when designing beam-tracking and link-budget fade allowances [22].

The fade depth due to scintillation for a specific time percentage p is commonly expressed as short-term amplitude fluctuations using the time percentage model (ITU-R P.618, Annex 1 [22]):

$$A_{\text{scint}}(p) = a(p) \cdot \sigma_{\text{pre}}(f, \theta, \text{climate}) \quad [\text{dB}], \quad (6)$$

where $a(p)$ maps standard-deviation values to time-percentage fades (ITU-R P.618 Annex 1 [22]) and σ_{pre} is calculated from the frequency, elevation angle θ and local turbulence parameters in the recommendation. We then scale by $a(p)$ to obtain fade depths for availability analyses and fade-allowance setting in adaptive systems. For 38 GHz at 11.3° elevation and 0.01

So total slant-path attenuation for a backhaul link may be written as

$$L_{\text{total}} = L_{\text{FSPL}} + A_{\text{rain}} + A_{\text{cloud/fog}} + A_{\text{gas}} + A_{\text{scintillation}}, \quad (7)$$

6) *Equation (7) — Total Backhaul Link Attenuation [22]*: This equation sums all atmospheric and propagation effects (Eqs. (1)–(6)) to compute the worst-case link budget for the HAPS-to-Gateway backhaul at 38 GHz. The breakdown is: FSPL (Eq. (1)) + rain (Eq. (3)) + fog (Eq. (4)) + gas (Eq. (5)) + scintillation (Eq. (6)), per ITU-R P.618 design guidance [22].

C. 2 GHz Service/Relay Link Model (UAV-to-User Air-to-Ground)

Service and relay links operate at 2–2.2 GHz (S-band per 3GPP TS 38.821), connecting ground users either directly to HAPS (service link, high elevation angle) or via UAV relays at 500 m altitude (relay link). Unlike the 38 GHz backhaul and feeder links, the 2 GHz S-band is resilient to atmospheric effects (fog, rain, gas absorption all negligible per ITU-R P.618) but highly sensitive to building blockage. This section models LoS probability, path loss, and multi-node interference.

1) *Equation (8) — Building-Blockage LoS Probability [18]*: For a UAV u at horizontal distance $r_{u,k}$ from user k , the LoS probability follows the exact building-blockage product series of [17], originally attributed to [18] (IEEE Trans. Antennas Propag., 2008, vol. 56, no. 4, pp. 1078–1084):

$$P_{u,k}^{\text{LoS}} = \prod_{n=0}^J \left[1 - \exp \left(-\frac{1}{2\xi^2} \left(h_u - \frac{n+\frac{1}{2}}{J+1} (h_u - h_k) \right)^2 \right) \right], \quad (8)$$

Exact product-series model [18] for line-of-sight probability accounting for cumulative building blockage along the path. The product iterates over building layers from ground to UAV height.

2) *Equation (9) — Number of Building Layers [18]*:

$$J = \left\lfloor \frac{r_{u,k}}{1000} \sqrt{\alpha\beta} - 1 \right\rfloor, \quad (9)$$

Computed from horizontal distance $r_{u,k}$ (meters), building density β [buildings/km²], and built-up ratio α [dimensionless] per Holis & Pechac Table I [18].

Parameters from Table I (Holis & Pechac 2008, IEEE Trans. Antennas Propag. [18]): $P_{\text{NLoS}} = 1 - P_{\text{LoS}}$; α is the built-up land ratio, β the mean building density [buildings/km²], and ξ a building-height parameter [m]. All are environment-dependent: for urban areas [18], $\alpha = 0.3$, $\beta = 500$, $\xi = 15$ m; dense urban uses $\alpha = 0.5$, $\beta = 300$, $\xi = 20$ m; suburban $\alpha = 0.1$, $\beta = 750$, $\xi = 8$ m; high-rise $\alpha = 0.5$, $\beta = 300$, $\xi = 50$ m (Table I) [18].

Landscape-based relay placement: The LoS probability model above (Eq. (8)–(9)), with its environment-dependent α, β, ξ parameter sets (urban, dense urban, suburban, high-rise), already provides the mechanism for determining where a UAV relay stays in LoS of its served users. This same mechanism determines the UAV relay’s altitude and horizontal-offset placement per landscape, as introduced in §II (Relay Link Geometry).

Single-relay scope: This document limits the topology to one UAV relay (Table I), so the interference sum in the SINR expression (Eq. (13), generalized as Eq. (39)) need only account for the 3 HAPS for the relay link – no UAV-to-UAV (relay-to-relay) interference term is needed.

3) *Equation (10) — Probability-Averaged Path Loss*: Unlike terrestrial base stations, HAPS nodes operate at high elevation angles. This near-zenith positioning provides a superior Line-of-Sight (LoS) probability, significantly reducing multi-path fading and shadowing in urban and mountainous terrain.

Each state $z \in \{\text{LoS}, \text{NLoS}\}$ adds an additional shadowing loss X_z in dB: the LoS term is zero-mean log-normal with $\sigma_{\text{LoS}} = 4\text{dB}$ (midpoint of the 3–5 dB range in Holis & Pechac [18]); the NLoS term has mean $\mu(\theta)$ and standard deviation $\sigma(\theta)$ from their Eq. (4) and Table IV (2 GHz, all environments), with θ the elevation of the UAV as seen from the user. Because the shadowing is log-normal in linear gain, the states are averaged in linear gain, not in dB:

$$\bar{g} = 10^{-\text{FSPL}/10} \left[P_{\text{LoS}} e^{\frac{(a\sigma_{\text{LoS}})^2}{2}} + P_{\text{NLoS}} e^{-a\mu(\theta) + \frac{(a\sigma(\theta))^2}{2}} \right], \quad (10)$$

$$\overline{\text{PL}} = -10 \log_{10} \bar{g}, \quad (11)$$

where $\text{FSPL} = 20 \log_{10}(4\pi f d_{3D}/c)$ and P_{LoS} is the LoS probability (Eq. (8)). The 3D distance is $d_{3D} = \sqrt{r^2 + (h_u - h_k)^2}$ where r is horizontal distance and h_u, h_k are UAV and user altitudes. At elevations where P_{LoS} is near 0 or 1 the linear and dB averages agree; in between they differ by up to about 17 dB, so the linear form is required for the expected received power.

The received power follows the standard link-budget relation

$$P_r = P_t + G_t + G_r - \text{PL}(d, \theta, \text{state}), \quad (12)$$

where P_t is transmit power, G_t and G_r are the transmit and receive antenna gains, and $\text{PL}(d, \theta, \text{state})$ is the path loss for the current geometry and propagation state. A blocked or NLoS link therefore yields a smaller P_r through a larger path-loss term (equivalently, a lower channel gain $g_{u,k}$); it is not added as a separate degradation term in the SINR expression itself. The reduced signal power is already embedded in the SINR numerator before interference and noise are considered.

The downlink SINR is

$$\gamma_{u,k} = \frac{p_u g_{u,k}}{\sum_{u' \neq u} p_{u'} g_{u',k} + \sigma_0^2}, \quad g_{u,k} = 10^{-\overline{\text{PL}}/10}, \quad (13)$$

4) *Equation (11) — Signal-to-Interference-plus-Noise Ratio (SINR)*: Computes downlink SINR at user k from UAV u per standard information-theory formulation. p_u is transmit power [Watts], $g_{u,k} = 10^{-\overline{\text{PL}}/10}$ is channel gain (reciprocal of path loss in linear scale), interference sums over competing UAVs u' , and σ_0^2 is thermal noise power [Watts]. For the noise-limited characterization below, interference is negligible; the characterisation below is noise-limited.

IV. SIMULATION AND RESULTS

All numerical results in this section were generated with a Python simulation stack: NumPy/pandas for the channel, battery, and association models, and Matplotlib for figure generation. Atmospheric and scintillation losses (rain, cloud/fog, gas, and ITU-R P.618 scintillation) are computed with the ITU-Rpy (itur) library, which implements the relevant ITU-R propagation recommendations directly rather than re-deriving them by hand. The learned association policy (LSTM and DRL) (Section V-B) uses a custom Deep Q-Network agent implemented in PyTorch (torch/torch.nn), rather than a third-party RL library,

trained against the per-user association environment described above. The corresponding data-generation and table/figure-generation scripts, along with the full numerical tables underlying the figures in this section, are provided in the Appendix.

A. Loss applicable to both frequencies

1) *Free-space path loss*: Both 2 GHz and 38 GHz bands are governed by the same universal equations for free-space path loss (Eq. (1))

$$a = \frac{\ln 10}{4\pi f} \text{FSPL} = 32.44 + 20 \log_{10}(f) + 20 \log_{10}(d)$$

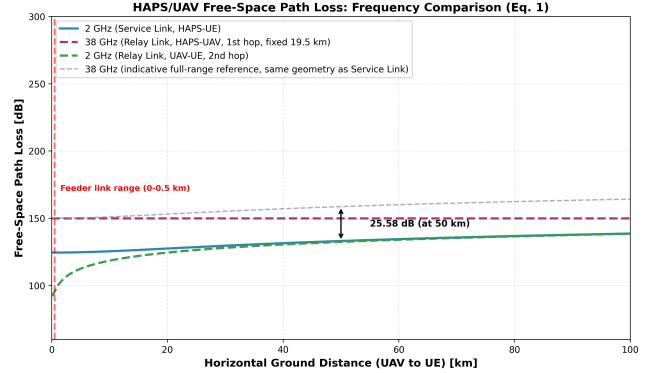


Fig. 2: HAPS/UAV free-space path loss, (1), over a 0–100 km horizontal footprint.

In the above figure we can see the FSPL for the following paths,

- Solid blue 2 GHz service link (HAPS→UE).
- Dashed green 2 GHz relay link, 2nd hop (UAV→UE).
- Dashed purple 38 GHz relay link, 1st hop (HAPS→UAV) this leg does not extend past the UAV; hence flat
- Dashed grey 38 GHz reference (HAPS→UE) for comparison only, no such direct link exists in the system model.

Observations:

- Every 10× increase in distance adds $20 \log_{10}(10) = 20\text{dB}$ path loss.
- A constant **25.6 dB offset** could be seen between 38 GHz and 2 GHz links $20 \log_{10}(38/2)$.
- FSPL is per-hop; the hops are independent RF links and hence bottlenecked by the weaker hop (min of per-hop capacities).

2) *Scintillation fade depth*: Tropospheric scintillation is rapid amplitude fluctuations caused by refractive-index turbulence in the lower atmosphere. The fade $A_s(p)$, exceeded for $p\%$ of the time, is computed with the ITU-R P.618 method (itur implementation, scintillation attenuation) at the site 28.61 N, 77.21 E, with $T = 15^\circ\text{C}$, relative humidity 60 %, pressure 1010 hPa and antenna efficiency $\eta = 0.5$. The aperture is $D = 0.31\text{ m}$ for the 38 GHz gateway link (the working-set gateway dish) and $D = 1\text{ m}$ for the 2.1 GHz link (a stated reference aperture). The table of $A_s(p)$ used in the model is:

Elevation	38 GHz, $D = 0.31$ m		2.1 GHz, $D = 1$ m	
	$p = 0.01$ %	$p = 1$ %	$p = 0.01$ %	$p = 1$ %
10°	1.76 dB	0.73 dB	0.33 dB	0.14 dB
30°	0.49 dB	0.21 dB	0.09 dB	0.04 dB
60°	0.25 dB	0.11 dB	0.05 dB	0.02 dB
89.86°	0.21 dB	0.09 dB	0.04 dB	0.02 dB

The fade at 30° is about 0.5 dB at 38 GHz and about 0.09 dB at 2.1 GHz for $p = 0.01$ %. The fade grows with frequency and with decreasing elevation, as the P.618 method predicts. The earlier empirical fit, with an exponent of 0.7 that was not validated by measurement, is no longer used.

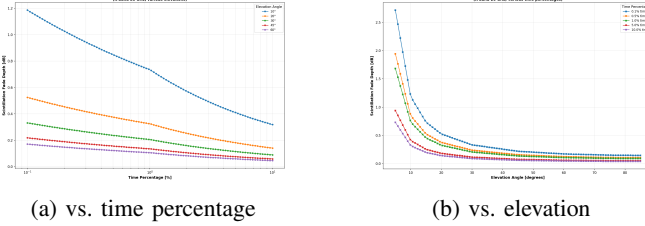


Fig. 3: Scintillation fade depth characteristics (ITU-R P.618, itur, gateway geometry at 38 GHz with $D = 0.31$ m and 2.1 GHz with $D = 1$ m): (a) Fade depth vs. time percentage at five elevation angles (10° – 60°). (b) Fade depth vs. elevation angle at four time percentages (0.01%–10%).

Observations:

- The backhaul/feeder links (38 GHz) carry a 25.6 dB FSPL penalty as well as 8.95 dB scintillation fade penalty.
- In general, the scintillation formula indicates fade depth increases with frequency, decreases with elevation angle.
- This favors positioning relay UAVs at higher elevation angles near the nadir when feasible, especially for the HAPS-UAV leg.

A Johnson–Nyquist thermal-noise model, $N = kTB$, or equivalently $N_{\text{dBm}} = -174 + 10 \log_{10}(B) + NF$, would provide a more physically grounded noise estimate than the assumed flat -100 dBm noise floor used elsewhere in this document. However, for short-range backhaul-link analysis, replacing the assumed noise floor with a bandwidth- and receiver-dependent thermal-noise value would primarily change the absolute SINR level rather than its trend with distance. A more detailed link-budget analysis incorporating the actual channel bandwidth, receiver noise figure, directional gateway antenna gain, and HAPS transmit power is deferred to the next phase.

B. Loss specific to 2GHz band

1) *SINR for different landscapes*: 2–2.2 GHz S-band used in service and relays are resilient to atmospheric effects (fog, rain, gas absorption all negligible per ITU-R P.618) however they are highly sensitive to building blockage, the following chart shows the path loss with 2GHz links in different landscapes.

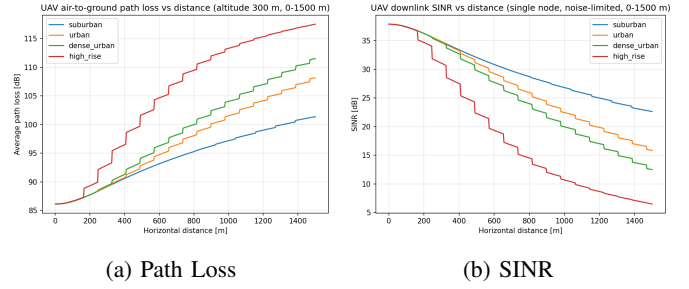


Fig. 4: 2 GHz service/relay link propagation (UAV altitude 500 m)

Observations:

- The Suburban landscape has lowest loss;
- High-rise landscape has higher blockage loss as well as sharp LoS/NLoS transitions.

C. Loss specific to 38 GHz Band

1) *Why 38 GHz instead of 2 GHz for the backhaul and feeder links*: At 38 GHz, $\lambda = 7.9$ mm, and the antenna gains of the link budget (§II-B) set the 3 dB beamwidths $\theta \approx \sqrt{41253/G}$ (degrees, G linear): about 2.1° for the gateway dish ($G_g = 39.7$ dBi, ≈ 0.31 m aperture at $\eta \approx 0.6$), about 5.9° for the UAV array ($G_u = 30.7$ dBi, ≈ 0.11 m), and about 41° for the HAPS ($G_h = 13.8$ dBi, a small patch). These narrow beams enable point-to-point backhaul/feeder links with interference rejection. Also higher frequencies enable wider bandwidth (Ka-band supports 1+ GHz channels). Also ITU-R allocates 2 GHz (S-band) for *terrestrial mobile access* (3GPP) and 38–39.5 GHz (Ka-band) for *fixed HAPS backhaul* (ITU-R M.2101). However 38 GHz band presents specific challenges with atmospheric loss, rain attenuation, and scintillation. The following sections analyze the frequency effects on atmospheric loss in detail.

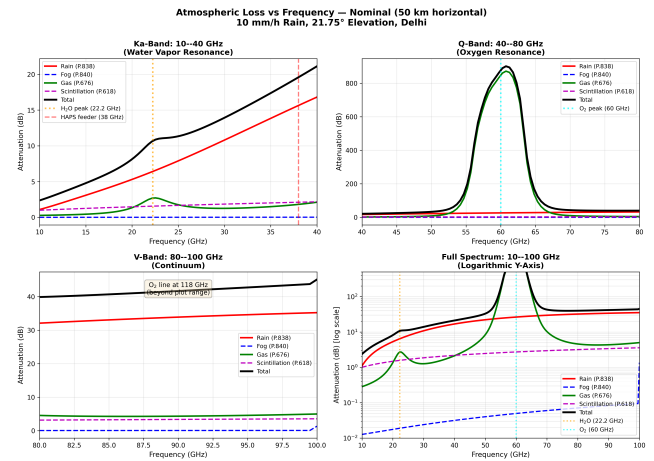


Fig. 5: Atmospheric loss vs. frequency.

Observation:

- **Frequency-dependent atmospheric loss**: Atmospheric attenuation varies strongly with frequency across the 10–

Component	Loss [dB]
Rain (P.618/P.838)	15.65
Fog/cloud (P.840)	0.02
Atmospheric gas (P.676)	1.81
Scintillation (P.618, 0.01% outage)	4.37
Total	21.85

TABLE III: Atmospheric loss breakdown, nominal case: 38 GHz backhaul link, 21.75° elevation, 54 km slant path, 10 mm/h rain, Delhi.

Component	Loss [dB]
Rain (P.618/P.838)	29.59
Fog/cloud (P.840)	0.04
Atmospheric gas (P.676)	3.43
Scintillation (P.618, 0.01% outage)	9.53
Total	42.59

TABLE IV: Atmospheric loss breakdown, worst case: 38 GHz backhaul link, 11.30° elevation, 102 km slant path, 10 mm/h rain, Delhi.

100 GHz range. The spectrum is characterized by distinct molecular absorption resonances rather than a uniform increase in loss with frequency.

- **Water-vapor resonance:** A pronounced atmospheric absorption peak occurs near 22.2 GHz due to the water-vapor resonance. This makes the region around 22.235 GHz unsuitable for a low-loss HAPS backhaul/feeder link.
- **Oxygen resonance:** A much stronger absorption peak occurs around 60 GHz due to molecular oxygen. The Q-band therefore contains a severe atmospheric-loss region that is undesirable for the backhaul/feeder links.
- **38 GHz backhaul/feeder-band location:** The selected 38 GHz HAPS backhaul/feeder frequency lies between the 22.2 GHz water-vapor and 60 GHz oxygen absorption peaks. It therefore avoids the two major molecular-resonance regions and lies in a comparatively lower-loss region of the spectrum.
- **Rain attenuation increases with frequency:** Rain attenuation increases with frequency over the considered 10–100 GHz range, as characterized by the frequency-dependent coefficients k and α in ITU-R P.838 [5].
- **Frequency-selection tradeoff:** Although the 2 GHz service-link frequency would provide lower atmospheric and rain attenuation, it is not directly suitable for the backhaul/feeder links because of regulatory, link-budget, antenna, and capacity constraints. The 38 GHz band therefore represents a practical compromise between propagation loss, available bandwidth, spectrum allocation, and antenna gain.
- **Effect of elevation angle and path length:** A lower elevation angle increases the slant-path length and consequently increases the magnitude of the attenuation across the spectrum.
- **Nominal versus worst-case geometry:** For the nominal case (50 km horizontal distance and 21.75° elevation), the

slant path is approximately 54 km, whereas the worst-case case (11.3° elevation and 100 km service-edge distance) has a slant path of approximately 102 km. The longer worst-case path therefore produces substantially higher atmospheric loss while preserving the same frequency-dependent resonance structure.

- **Overall backhaul/feeder-frequency implication:** The spectrum analysis indicates that backhaul/feeder-frequency selection cannot be based solely on minimizing frequency-dependent rain attenuation. The selected frequency must simultaneously avoid severe molecular absorption bands and provide adequate spectrum and antenna characteristics. The 38 GHz band satisfies this tradeoff by operating between the major water-vapor and oxygen absorption resonances.

D. Loss specific to Multi-HAPS interference

The multi-HAPS interference mechanism is fundamentally a spatial phenomenon. A serving HAPS provides the desired signal, while the remaining HAPS transmit on the same radio resource and contribute co-channel interference at the same receiver:

$$\text{HAPS}_0 \rightarrow \text{Receiver (desired)} \quad (14)$$

while

$$\text{HAPS}_1, \text{HAPS}_2, \dots \rightarrow \text{same receiver (interference)}. \quad (15)$$

The resulting SINR therefore depends not only on the horizontal distance from the serving HAPS nadir, but also on the angular position of the receiving terminal relative to the interfering HAPS. This dependence is particularly important for the 2 GHz service/relay footprint, where a receiver can lie within the common coverage region of several HAPS.

1) *General Multi-HAPS SINR Formulation:* The backhaul-link, feeder-link (HAPS-to-UAV), and HAPS-to-UE service-link SINR expressions developed below are each instances of a common HAPS-to-receiver SINR model. For a serving HAPS h and an arbitrary receiving node r , the SINR is

$$\gamma_{h,r} = \frac{P_h G_h G_r g_{h,r}}{N_r + \sum_{h' \neq h} P_{h'} G_{h'} G_r g_{h',r}} \quad (16)$$

where r can represent either a Gateway or a UAV.

The numerator represents the desired HAPS transmission, the first term in the denominator represents receiver thermal noise, and the summation represents aggregate co-channel interference from the remaining HAPS.

In the linear domain, the channel gain is obtained from the total propagation loss as

$$g_{h,r} = 10^{-L_{h,r}^{\text{tot}}/10} \quad (17)$$

Equivalently, the SINR in decibels is

$$\text{SINR}_{\text{dB}} = 10 \log_{10} \left(\frac{P_{\text{desired}}}{P_{\text{noise}} + P_{\text{interference}}} \right) \quad (18)$$

It is important that the propagation model and the interference model remain conceptually separate. FSPL and atmospheric/scintillation losses determine the channel gain of each desired and interfering link, whereas the SINR equation combines those received powers with receiver noise and co-channel interference.

2) *HAPS-to-Gateway Backhaul Link*: The backhaul link connects each HAPS to its associated ground gateway and operates at 38 GHz. The desired received signal is generated by the serving HAPS, while transmissions from other HAPS constitute potential co-channel interference if the same time-frequency resource is reused. For a serving HAPS h and gateway g , the received desired power is

$$P_{h,g}^{\text{rx}} = P_h G_h G_g g_{h,g} \quad (19)$$

where P_h is the HAPS transmit power, G_h and G_g are the transmit and receive antenna gains, respectively, and $g_{h,g}$ is the channel gain. The baseline propagation loss is modeled using free-space path loss (FSPL):

$$L_{\text{FSPL}}(h, g) = 20 \log_{10} \left(\frac{4\pi d_{h,g}}{\lambda} \right) \quad (20)$$

where $d_{h,g}$ is the HAPS-to-Gateway slant distance and λ is the wavelength.

When atmospheric impairments are included, the total propagation loss is written as

$$L_{h,g}^{\text{tot}} = L_{\text{FSPL}} + A_{\text{rain}} + A_{\text{cloud/fog}} + A_{\text{gas}} + A_{\text{scint}}, \quad (21)$$

with all terms expressed in dB. The corresponding linear channel gain is

$$g_{h,g} = 10^{-L_{h,g}^{\text{tot}}/10}. \quad (22)$$

The backhaul-link SINR is therefore

$$\gamma_{h,g} = \frac{P_h G_h G_g g_{h,g}}{N_g + \sum_{h' \neq h} P_{h'} G_{h'} G_g g_{h',g}}, \quad (23)$$

where N_g is the gateway receiver noise power.

The summation term represents potential multi-HAPS interference. It should not be interpreted as necessarily significant for every backhaul-link geometry. In the present constellation, the gateway is located close to the nadir of its serving HAPS, whereas the other HAPS are approximately 65 km away from the serving HAPS. Consequently, their 38 GHz signals experience substantially greater propagation loss than the desired backhaul signal. Under the present power, antenna, and geometry assumptions, this interference is expected to be negligible compared with the receiver noise, resulting in a predominantly noise-limited backhaul link.

Thus, multi-HAPS interference is *modeled as a potential impairment* on the backhaul link, but the present configuration produces a negligible interference contribution. This distinction is important because backhaul-link interference can become relevant in a different frequency-reuse pattern, antenna pattern, gateway placement, or constellation geometry.

3) *HAPS-to-UAV Relay First Hop (Feeder Link)*: The second HAPS-related link is the feeder link, the first hop of the relay path,

$$\text{HAPS} \rightarrow \text{UAV} \rightarrow \text{UE}. \quad (24)$$

For the first hop, the UAV acts as the receiving node. The serving HAPS transmits the desired signal, while the remaining HAPS can simultaneously transmit on the same channel and therefore act as co-channel interferers.

For serving HAPS h and receiving UAV u , the desired received power is

$$P_{h,u}^{\text{rx}} = P_h G_h G_u g_{h,u}. \quad (25)$$

The corresponding multi-HAPS SINR is

$$\gamma_{h,u} = \frac{P_h G_h G_u g_{h,u}}{N_u + \sum_{h' \neq h} P_{h'} G_{h'} G_u g_{h',u}}, \quad (26)$$

where N_u is the thermal noise power at the UAV receiver.

This is the primary multi-HAPS interference scenario considered in the relay architecture. The interference term explicitly represents simultaneous transmissions from the non-serving HAPS to the same UAV receiver.

The channel gain for every HAPS-to-UAV path is obtained from its corresponding propagation loss:

$$g_{h,u} = 10^{-L_{h,u}^{\text{tot}}/10}. \quad (27)$$

The total loss may be represented as

$$L_{h,u}^{\text{tot}} = L_{\text{FSPL}}(h, u) + A_{\text{atm}}(h, u) + A_{\text{scint}}(h, u), \quad (28)$$

where A_{atm} represents the relevant atmospheric attenuation components and A_{scint} represents tropospheric scintillation when included in the channel realization.

Unlike the low-altitude UAV-to-UE link, the HAPS-to-UAV path is a high-altitude slant path and is therefore dominated by the geometry between the stratospheric HAPS and the receiving UAV. The same propagation model must be applied independently to the desired HAPS and each interfering HAPS so that their different slant ranges and elevation angles are reflected in the resulting SINR.

4) *UAV-to-UE Relay Second Hop*: The second relay hop is

$$\text{UAV} \rightarrow \text{UE}. \quad (29)$$

This link is not itself a multi-HAPS interference link because the transmitter is the UAV rather than the HAPS. Its

SINR should therefore be modeled using the relevant UAV transmitters and receivers:

$$\gamma_{u,k} = \frac{P_u G_u G_k g_{u,k}}{N_k + \sum_{u' \neq u} P_{u'} G_{u'} G_k g_{u',k}}, \quad (30)$$

where k denotes the UE and u' denotes other simultaneously transmitting UAVs.

If only one UAV is transmitting in the considered resource block, the interference summation is zero and the link becomes noise-limited. If multiple UAVs reuse the same channel, their transmissions constitute the interference term.

E. Relay Path Analysis (HAPS $\rightarrow$ UAV $\rightarrow$ UE)

The two SINR expressions above (Eq. (27), (31)) define the relay link; this section evaluates them numerically as a decode-and-forward (DF) two-hop path, implemented in `uav_relay_two_hop_capacity_bps_hz()`. The first hop (the feeder link) reuses the 38 GHz backhaul-link slant-path model (§III.B; UAV hovering at 500 m, ITU-P.618 scintillation included) since both the backhaul link and this hop are high-altitude, high-frequency, 38 GHz links; the second hop reuses the 2 GHz landscape-dependent LoS/NLoS blended model (§III.C) since the UAV, like the HAPS, is a low-altitude access node subject to building blockage. The two hops are combined DF, so the end-to-end capacity is the bottleneck of the two:

$$C_{\text{uav-relay}} = \min(C_{\text{haps-uav}}, C_{\text{uav-ue}}). \quad (31)$$

Minimum design SINR. A link is only useful if it clears a minimum design SINR set by the target block-error rate (BLER) of the modulation and coding scheme (MCS) it needs to run, plus an implementation margin:

$$\gamma_{\text{min,design}} = \text{SINR}_{\text{MCS}}(\text{target BLER}) + \text{margin}. \quad (32)$$

For the direct HAP $\rightarrow$ vehicle-UE service link, a lower bound of $\gamma_{\text{SINR}} = -5$ dB is used as the minimum link quality for a UE to be considered active and able to receive signals from the HAP – close to the ≈ -6 dB required by the lowest-order 3GPP NR MCS (QPSK, CQI index 1, BLER = 10%) [13]. The 38 GHz feeder hop is held to a higher design floor: relaying is only worth its extra hop and extra hardware if it can run at least a moderate-order MCS (16-QAM, CQI ≈ 7 , $\text{SINR}_{\text{MCS}} \approx +5$ dB at 10% BLER [13]), so with a 3 dB implementation margin this document targets $\gamma_{\text{min,design}}^{\text{haps-uav}} \approx +8$ dB for the HAPS $\rightarrow$ UAV hop.

Table XIII (§D) sweeps the HAPS–UAV horizontal distance 0–100 km across all four landscapes, and Figure 6 plots the resulting SINR on each hop against the direct HAPS $\rightarrow$ UE link.

Key finding: the relay outperforms the direct link at every distance, because co-channel interference from the other HAPS limits the direct service link. The UAV hovers directly above the user it serves (§II), so its horizontal distance from the user is small and fixed, while its distance from the serving HAPS grows with the user’s own distance from the HAPS

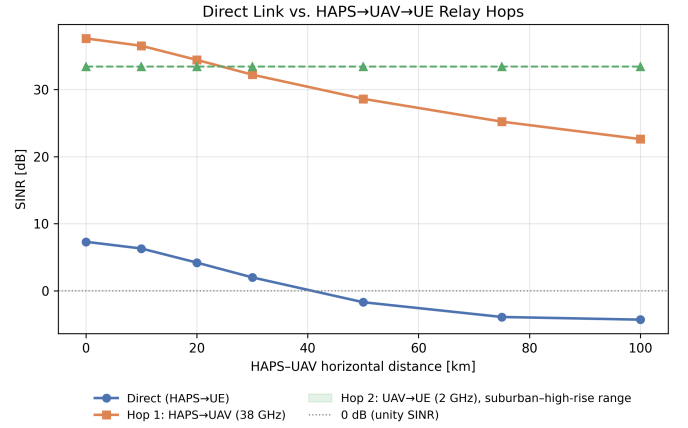


Fig. 6: Direct HAPS $\rightarrow$ UE SINR (suburban, averaged in dB over 24 azimuths, including co-channel interference from the other two HAPS) vs. the two UAV relay hops, as a function of HAPS–UAV horizontal distance. The HAPS $\rightarrow$ UAV feeder hop (orange, including interference from the other HAPS, which is about 35 dB below the noise floor) starts at +37.6 dB at nadir and stays above +22 dB at 100 km, well above the +8 dB design floor (Eq. (33)). The UAV $\rightarrow$ UE access hop (green, suburban and high-rise identical at 500 m) is +33.4 dB. The direct link falls from +7.3 dB at nadir to -9.1 dB at 100 km.

nadir. The UAV $\rightarrow$ UE hop therefore stays at $\text{SINR} \approx +33$ dB regardless of where the UAV is deployed (Table XIII). The HAPS $\rightarrow$ UAV hop is a 38 GHz slant path with antenna gains $G_h = 13.8$ dBi (HAPS side) and $G_u = 30.7$ dBi (UAV side), matching a published 30 GHz HAPS backhaul-class budget and a UAV-mountable mmWave patch-array antenna [14], [15]. Its SINR is +37.6 dB at nadir and falls to +22.6 dB at 100 km as the elevation angle drops to 11° . Its interference term is about 35 dB below the noise floor, so the hop is noise-limited. The direct HAPS $\rightarrow$ UE link is different: it shares the 2 GHz band with the other two HAPS, and its SINR falls from +7.3 dB at nadir to -9.1 dB at 100 km. Its capacity therefore drops from 2.7 to 0.2 bits/s/Hz. The relay’s end-to-end capacity is $C_{\text{uav-relay}} = \min(C_{\text{haps-uav}}, C_{\text{uav-ue}})$: 11.1 bits/s/Hz at nadir, where the UAV $\rightarrow$ UE access hop is the bottleneck, and 7.5 bits/s/Hz at 100 km, where the feeder hop is the bottleneck (from about 30 km outward). This result depends on two modelling choices: the 38 GHz scintillation fade from the ITU-R P.618 table (about 0.2 dB at zenith), and the ITU-R F.699 sidelobe envelope used for the interferers and the receive patterns (Section III, interference model). The DQN association agent (§V-B) does not route traffic through the UAV relay; adding the relay to its action space is noted as future work in the Conclusion (§VII).

1) *HAPS-to-UE Service Link:* The multi-HAPS service link is

$$\text{HAPS} \rightarrow \text{UE}. \quad (33)$$

Consider a three-HAPS equilateral constellation with serving HAPS S at the origin and interfering HAPS I_1 and I_2 separated from S by approximately 65 km. Let a receiver be located at horizontal distance r from the serving-HAPS nadir and at azimuth angle θ . The horizontal position of the receiver may be written as

$$\mathbf{x}_r = \begin{bmatrix} r \cos \theta \\ r \sin \theta \end{bmatrix}. \quad (34)$$

For an interfering HAPS i located at azimuth θ_i and horizontal separation D from the serving HAPS, the corresponding slant range is

$$d_i(r, \theta) = \sqrt{h_i^2 + r^2 + D^2 - 2rD \cos(\theta - \theta_i)}, \quad (35)$$

where h_i represents the effective vertical separation between HAPS i and the receiver. The desired slant range is

$$d_s(r) = \sqrt{h_s^2 + r^2}. \quad (36)$$

Thus, even when two receivers have the same radial distance r , their desired and interfering received powers can be different because their angular positions relative to I_1 and I_2 are different.

The angular separation between the desired-HAPS direction and an interfering-HAPS direction, as viewed from the receiver, is the angle subtended by the two HAPS at the receiver. It can be defined from the corresponding line-of-sight vectors as

$$\Delta\phi_i = \cos^{-1} \left(\frac{(\mathbf{p}_s - \mathbf{x}_r) \cdot (\mathbf{p}_i - \mathbf{x}_r)}{\|\mathbf{p}_s - \mathbf{x}_r\| \|\mathbf{p}_i - \mathbf{x}_r\|} \right), \quad (37)$$

where $\mathbf{p}_s$ and $\mathbf{p}_i$ denote the three-dimensional positions of the serving and interfering HAPS, respectively. Thus, $\Delta\phi_i$ is the actual angular separation between the desired and interfering directions at the receiver. In the present free-space propagation model, this angle does not by itself produce constructive or destructive combining; rather, the receiver position changes the corresponding slant ranges and hence the desired and interfering received powers. If directional receive or transmit beam patterns are introduced, the same angular separation can additionally affect the antenna gains toward the serving and interfering HAPS.

The multi-HAPS SINR can therefore be viewed as a spatial function

$$\gamma(r, \theta) = \frac{P_s G_s G_r g_s(r)}{N_r + \sum_{i \in \mathcal{I}} P_i G_i G_r g_i(r, \theta)}, \quad (38)$$

where $\mathcal{I}$ denotes the set of interfering HAPS. Because $g_i(r, \theta)$ changes with the receiver position, the SINR forms a spatial map rather than a function of distance alone.

a) Good and bad reception regions.: The original service-link results demonstrate this spatial behavior directly. At a horizontal distance of $r = 50$ km, the SINR varies from approximately -7.40 dB when the receiver is oriented toward interferer I_1 ($\theta = 0^\circ$) to approximately $+2.54$ dB when positioned away from the constellation ($\theta = 210^\circ$), giving a spatial spread of approximately 9.94 dB. The corresponding constellation-sector representation identifies a good sector approximately in the 120° – 290° region, away from the interfering HAPS, and a bad sector approximately in the 330° – 75° region, toward the interferers.

These regions should be interpreted as regions of higher and lower SINR, respectively. In the good sector, the interfering HAPS contribute less received power relative to the serving HAPS, reducing the interference term. In the bad sector, one or more interfering HAPS have a more favorable geometry to the receiver, increasing their received interference power and reducing the SINR. Hence, multi-HAPS interference produces a spatially non-uniform service area even when the receiver is at the same radial distance from the serving HAPS.

This observation is also reflected in the min-max angular spread of the SINR results. The angular variation becomes increasingly important with distance, so a distance-only link budget is insufficient to characterize the interference environment. The receiving-terminal position relative to the constellation must be retained explicitly in the multi-HAPS model.

Figure 7 shows this constellation geometry directly, together with the resulting spatial SINR pattern it produces at $r = 50$ km: a good sector (away from both interferers) and a bad sector (toward them), rather than a distance-only falloff.

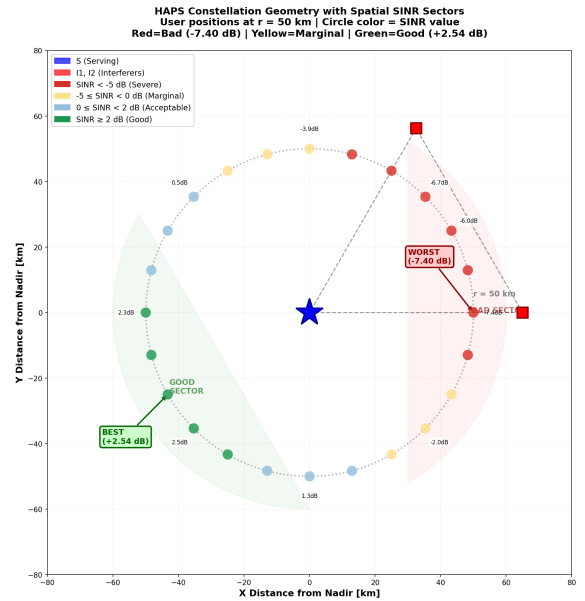


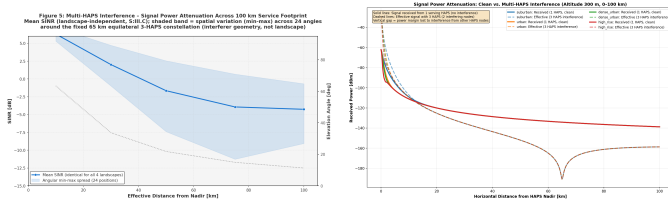
Fig. 7: HAPS constellation geometry (true 65 km equilateral triangle: S at origin, I_1 at 0° , I_2 at 60°) with spatial SINR sectors at $r = 50$ km. Good sector (green, away from interferers, $\sim 120^\circ$ – 290°) vs. bad sector (red, toward interferers, $\sim 330^\circ$ – 75°). Best $+2.54$ dB @ 210° ; worst -7.40 dB @ 0° .

2) *Multi-HAPS Interference Conclusion:* Good/bad sector result: At $r = 50$ km, SINR ranges from -7.40 dB toward interferer I_1 ($\theta = 0^\circ$) to $+2.54$ dB away from the constellation ($\theta = 210^\circ$), a 9.94 dB spread. This maps to a good sector (approximately 120° – 290° , away from interferers) and a bad sector (approximately 330° – 75° , toward them), as shown in Figure 7.

So multi-HAPS interference makes the service area spatially non-uniform even at fixed radial distance r .

The same interference term exists in the general SINR formula for the 38 GHz backhaul link, but under the current geometry (gateway near serving-HAPS nadir, interferers 65 km away) that interference is expected to be negligible vs. receiver noise. Bottom line: the service-link analysis is scoped to the angularly-varying interference pattern (good/bad sectors, SINR spread), while the backhaul link is reported as predominantly noise-limited under this constellation/gateway placement.

a) *Multihaps validation from simulation:* Figure 8 quantifies the interference-induced SINR penalty across the 100 km service footprint, averaged over all 24 angular positions around the constellation (all four landscapes are numerically identical at this tier, so a single line is shown rather than four overlapping ones). The shaded band is the min–max spread from that angular variation, not measurement uncertainty. Mean SINR falls from $+6.3$ dB at 10 km to -4.3 dB at 100 km, while the band widens from 1.95 dB at 10 km to 11.94 dB at 75 km – beyond ~ 30 km, angular position relative to the 65 km equilateral constellation dominates over distance decay alone. In the received-power trace, the downward bend after ~ 60 km marks the transition from an interference-limited to a path-loss-dominated regime: desired power falls faster than interference, so the absolute signal level keeps decaying even once interference stops being the dominant cause of degradation. The network must therefore employ TDMA, FDMA, beamforming, or power control to manage this interference and its spatial variability.



(a) Mean SINR and angular min–max spread across 100 km service footprint. (b) Received-power comparison showing clean vs. effective power under three-HAPS interference.

Fig. 8: Multi-HAPS interference characterization at the service link: (a) SINR variability and (b) received-power degradation.

Figure 9 decomposes this along one representative direction into SINR_Single (no interference), SINR_Actual (with 3-node interference), and SINR_Interferers (interference-to-noise ratio), computed from Equation (39) using pure free-space path loss. SINR_Interferers stays positive (0.95–4.6 dB)

across the full 10–100 km range, so the link never leaves the interference-limited regime; SINR_Actual itself only turns negative between 30 km ($+2.0$ dB) and 50 km (-1.6 dB), reaching -4.3 dB at 100 km – because path loss to the serving HAPS eventually dominates the link budget, not because the interference term grows.

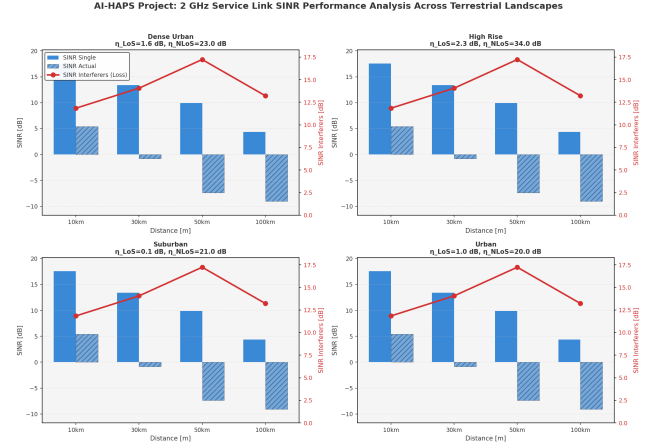


Fig. 9: 2 GHz service link SINR, four identical landscape subplots (landscape-independent, §III.C), 10–100 km distances.

Figure 10 shows the 38 GHz backhaul-link SINR for gateway offsets of 10–500 m from nadir (gateway at 50 m altitude, HAPS at nominal stratospheric altitude), including the HAPS-side and Gateway-side antenna gains of §II-B ($G_h = 13.8$ dBi, $G_g = 39.7$ dBi). The backhaul slant range is set by the HAPS altitude (~ 20 km), against which a 10–500 m horizontal offset is negligible, so SINR is essentially flat at $\approx +38.0$ – 38.1 dB across this range – strongly positive once the directional antenna gains are included, versus the deeply negative result obtained when the link budget used raw transmit power alone (§II-B). The interfering HAPS, by contrast, sit at a slant range of ~ 65 – 68 km (equilateral-triangle geometry, §III.A); the resulting interference power is ≈ -35 dB below the noise floor. This assumes that each interferer’s main beam points at its own service area, so the gateway sees it through the sidelobe envelope of ITU-R F.699 (floored at -10 dBi), and that the gateway dish is off-boresight by about 73° toward the interferer, so its gain there is also the -10 dBi floor. The backhaul link is therefore **noise-limited, not interference-limited**, unlike the service link.

3) *Service Link (2 GHz): SINR Effects from Distance and Constellation Geometry:* In summary, 2 GHz service-link SINR is governed by horizontal distance r from the HAPS nadir and by the receiver’s angular position relative to the two interfering HAPS in the 65 km equilateral-triangle constellation (§IV-D, Figure 7); the landscape environment has no effect, since both the desired and interfering links use free-space path loss. Distance alone is therefore insufficient to characterize interference conditions, motivating the DRL-based adaptive scheduling approach that follows.

AI-HAPS Project: 38 GHz Feeder Link (HAPS-to-Gateway) Performance Analysis

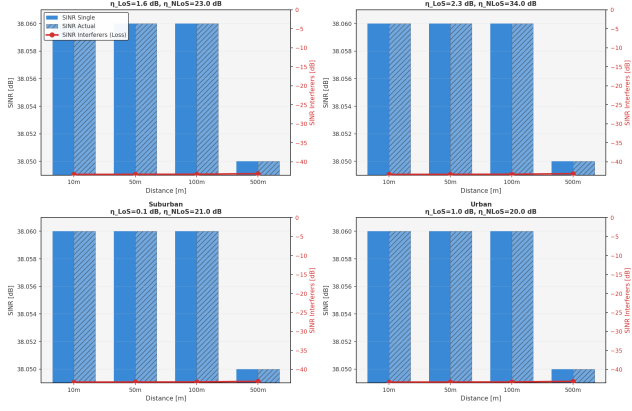


Fig. 10: 38 GHz backhaul link (HAPS-to-Gateway) SINR, four identical landscape subplots (landscape-independent, §III.C), 10–500 m distances.

F. DRL Comparison with Per-User State

Building on the 3-HAPS constellation (Table I) and the SINR model of Sections III.A–III.C, this section defines the per-user HAPS-association task used to compare closed-form and learned (DRL) policies: at each simulated hour, every one of $N = 50$ users must choose which of the 3 HAPS to associate with, based on a per-user state consisting of its SINR to each of the 3 HAPS (dB, Eq. (13)), its horizontal distance to each (km), each HAPS’s battery state of charge (%), and the current weather (rain rate, visibility, hour of day). One episode is one simulated day (24 hourly steps). Per episode, user positions and each HAPS’s initial battery SoC ($\mathcal{U}(30, 100)$ %) are drawn once and held fixed; weather is drawn once per episode (WeatherScenarioGenerator, random_mix mode). Each HAPS’s battery is charged by a fixed diurnal solar profile and discharged by a constant platform/backhaul draw plus its currently-associated users’ share of load ($P_{k,\text{user}} = 1$ W per user, ASSUMED); the solar/platform terms dominate, so time-of-day remains the primary driver of SoC and the association policy has a secondary, but non-trivial, effect on it. Rain de-rates only the backhaul-link throughput credit (Section IV-D), not the 2 GHz service SINR.

Two closed-form (non-learned) association policies are evaluated here:

- **Random:** uniform-random platform choice each hour (sanity floor).
- **Max-SINR:** each user k always associates to the platform

$$p^* = \arg \max_p \text{SINR}_{k,p}, \quad (39)$$

- pure closed-form physics, no memory of battery or fairness state, no learning.

Both are evaluated over 50 held-out episodes (seed 999) using Jain’s fairness index, outage fraction (SINR below -2 dB, the severe-outage threshold measured in Section IV.B/Table A.3),

TABLE V: Association baseline results: 50 held-out 24-hour episodes (seed 999), 50 users, mean $\pm$ std

Policy	Jain	Outage /50	Throughput [bits/s/Hz]
Random	0.331 ± 0.046	33.7 ± 3.3	33.95 ± 4.75
Max-SINR	0.700 ± 0.027	1.0 ± 0.9	76.18 ± 3.57

and aggregate throughput (Shannon capacity, summed over users) as metrics. Jain’s fairness index measures how evenly a resource (here, per-user throughput T_k) is shared across the N users,

$$J(\mathbf{T}) = \frac{\left(\sum_{k=1}^N T_k\right)^2}{N \sum_{k=1}^N T_k^2}, \quad J \in \left(\frac{1}{N}, 1\right], \quad (40)$$

with $J = 1$ when all users receive identical throughput (perfectly fair) and $J \rightarrow 1/N$ when the resource is concentrated on a single user (maximally unfair); it is reported alongside outage and aggregate throughput because a policy that maximizes total throughput can still do so unfairly, e.g. by starving low-SINR users. Section IV-G reports the resulting comparison; Section V-B (the learned association policy) later adopts this same per-user state and evaluates whether a learned policy can improve further on the closed-form baseline established there.

G. Multi-HAPS Association Baseline Results (Random vs. Max-SINR)

Using the setup of Section IV-F, this measures the two closed-form association policies against each other. All numbers are measured over 50 held-out 24-hour episodes (seed 999), 50 users.

Even this simple, closed-form per-user rule – always associate to whichever HAPS gives the best instantaneous SINR – more than doubles Jain fairness, cuts outage by $34\times$ (33.7 to 1.0 of 50 users), and more than doubles aggregate throughput relative to random association. This confirms, at the level of a full multi-user network simulation, what the spatial SINR heterogeneity in Table A.3 already implied: association choice matters far more than any fixed or naive assignment. Section V-B (the learned association policy) evaluates whether a learned policy can improve further on this closed-form baseline.

V. LSTM AND DRL FOR USER–PLATFORM ASSOCIATION

This section combines two learned components for dynamic user-to-platform association in the HAPS–UAV–Gateway network. An LSTM forecasts the feeder rain attenuation up to six hours ahead from past weather, and a Deep Q-Network (DQN) agent uses its one-hour-ahead forecast to choose each user’s platform, maximizing throughput, fairness, and energy efficiency. Whereas Phase 1 characterized static link quality, this section learns real-time association decisions.

A. LSTM Rain-Attenuation Forecaster

a) *Why an LSTM is needed.*: The DQN chooses an association at each hour t , before that hour’s rain is observed. The ITU-R P.618 attenuation [22] can be evaluated only for

TABLE VI: LSTM against persistence on 2026 (6557 forecast hours, no 2026 data in training): mean absolute error (MAE) and root-mean-square error (RMSE), in dB, per forecast horizon. Bold marks the better method for each measure and horizon.

Horizon	LSTM MAE	Persistence MAE	LSTM RMSE	Persistence RMSE
+1 h	0.20	0.12	0.65	0.53
+2 h	0.27	0.20	0.86	0.75
+3 h	0.31	0.26	0.96	0.90
+4 h	0.33	0.30	1.00	1.02
+5 h	0.34	0.34	1.00	1.11
+6 h	0.35	0.36	1.02	1.17

hours already observed, so the agent needs a forecast of the feeder loss for hour t . A long short-term memory (LSTM) network learns the mapping from the recent weather history to the next six hourly attenuation values, and the results below report its accuracy at each horizon. The agent uses only the one-hour-ahead output (the forecast for hour t , issued at hour $t-1$); the longer horizons are reported to show the forecast's skill, not as an input to the DQN. Persistence, which repeats the last observed value, is the benchmark the forecast must beat to justify its use.

b) How it is used.: The pipeline has four steps:

- 1) **Data preparation.** Hourly NASA POWER and METAR observations and ERA5 cloud fields for Delhi (2023–2026) are merged on UTC hours. The 14 input features are temperature, humidity, pressure, wind, precipitation, cloud amount, visibility, rain and fog codes, cloud liquid water, and sine and cosine encodings of time of day and season.
- 2) **Target.** For each hour, the target is the rain attenuation on the HAPS-to-gateway path at 38.0 GHz and 89.86° elevation, from the observed precipitation: γ_R from ITU-R P.838 at the hourly rate, multiplied by the slant length through the rain layer (P.839 rain height of 5.28 km at Delhi) and the P.618 path reduction factor. The same model gives the DQN's rain loss (Section VI-B).
- 3) **Windows.** Each sample uses the past 24 hours of features and predicts the next six hourly targets.
- 4) **Rolling episodes.** The DQN is trained on 400 rolling 24-hour episodes drawn from 2024, each starting at a distinct hour. At every hour of an episode, the LSTM forecast from the preceding 24 hours is supplied as the state feature.

Forecasts are always out of sample. The 2024 forecasts come from a model trained on 2023 only, the 2025 forecasts from a model trained on 2023–2024, and the 2026 forecasts from a model trained on 2023–2025. Table VI and Figure 11 report the 2026 validation, which uses no 2026 data in training.

c) Reading the validation.: Persistence is more accurate at +1 to +3 h on both measures. At +4 h the LSTM has the lower RMSE, and at +5 and +6 h it matches or beats persistence on MAE and RMSE. The gain over persistence is therefore modest and confined to the longer horizons. In rain hours (1248 of the 6557 forecasts) the LSTM is less

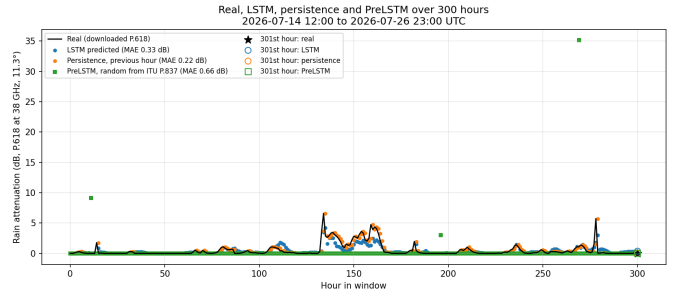


Fig. 11: Real, LSTM, persistence and PreLSTM over a 300-hour window in July 2026, the window with the most rain. Real: observed hourly attenuation from the downloaded precipitation (P.838 and P.618 method, gateway geometry). LSTM: one-hour-ahead forecast. Persistence: previous hour's value. PreLSTM: random hourly rain rates drawn from the ITU-R P.837 statistics for Delhi, converted with P.618; this is a synthetic baseline with no observed data. Star and open markers show the 301st hour.

accurate than persistence at +1 h (MAE 0.71 against 0.51 dB). Each method forecasts every hour of 2026, and the errors are averaged per horizon over the year. Persistence serves as the benchmark only. The DQN uses the one-hour-ahead forecast at every hour. Improving the forecast with current observations from neighbouring regions is future work.

B. Motivation: Why DRL?

The spatial SINR heterogeneity discovered in Phase 1 (9.94 dB variation at 50 km driven by constellation geometry alone, Figure 8) demonstrates that distance-only scheduling heuristics fail. A simple “always pick max SINR” rule cannot anticipate or adapt to the angular SINR patterns created by the 3-HAPS triangle. Moreover, when weather degrades feeder links (Section IV-D), the network must dynamically fail-over users from one platform to another while respecting battery constraints on each HAPS node.

DRL offers a principled approach: the agent observes the current network state (SINR to each platform, battery levels, weather forecast) and learns a policy $\pi(a|s)$ that selects the best platform for each user, maximizing cumulative reward. By training on diverse scenarios (rain events, user mobility, multi-node interference), the learned policy discovers non-obvious patterns (e.g., preemptively rerouting from a well-positioned but rain-threatened link to a less ideal but stable backup) that no hand-crafted rule would capture.

C. Per-User Markov Decision Process (MDP) Formulation

The model has 4 platform choices (HAPS_1, HAPS_2, HAPS_3, UAV_relay). With N users, a centralized MDP would face 4^N possible joint assignments—intractable for large networks. We instead decompose into per-user MDPs: each user independently chooses a platform, and the network computes cumulative reward after all assignments.

1) *State Space*: The state vector for user k is 12-dimensional:

$$\mathbf{s}_k = [\text{SINR}_{k,1,2,3}, r_{k,1,2,3}, \text{SoC}_{1,2,3}, R_{\text{rain}}, V_{\text{vis}}, t_{\text{hour}}] \quad (41)$$

Feature Definitions (cross-referenced to Phase 1 results):

- $\text{SINR}_{k,p}$: Downlink SINR at user k from platform p (dB). Phase 1 measured $[-7.4, +7.3]$ dB across 100 km footprint (Fig. 8); bounds $[-20, +30]$ dB provide headroom for edge cases.
- $r_{k,p}$: Horizontal distance user k to platform p (km). Range $[0, 100]$ km covers service footprint (§ II).
- SoC_p : Battery state-of-charge platform p (%). Observed by all users; range $[0, 100]$ % (empty to full).
- R_{rain} : Rainfall rate (mm/h). Range $[0, 50]$ mm/h (rare events).
- V_{vis} : Visibility (m). Range $[0, 10]$ km (fog/clear).
- t_{hour} : Time-of-day (normalized). Range $[0, 1]$ maps $0 \rightarrow$ midnight, $0.5 \rightarrow$ noon.

Observation Space Bounds (negative SINR preserved for outage signals):

$$\mathcal{S} = \underbrace{[-20, +30]^3}_{\text{SINR}_{k,p}} \times \underbrace{[0, 100]^3}_{r_{k,p} \text{ (km)}} \times \underbrace{[0, 100]^3}_{\text{SoC}_p \text{ (\%)}} \times \underbrace{[0, 50]}_{\text{rain}} \times \underbrace{[0, 10k]}_{\text{vis (m)}} \times \underbrace{[0, 1]}_{t} \quad (42)$$

DQN Scope (scintillation deferred to future work):

- **Included**: Deterministic SINR from constellation geometry (Phase 1); rainfall & visibility for backhaul fade planning; battery state for load balancing
- **NOT included**: Tropospheric scintillation fading (Section IV-A2) — time-varying random process requiring LSTM forecasting (future work)
- **Approximation**: the DQN does not model scintillation; the P.618 fade at $p = 1\%$ is 0.04 dB at 2.1 GHz and 0.6 dB at 38 GHz (Section IV-A2), which is not a DQN input
- **LSTM forecaster**: predicts rain attenuation up to six hours ahead from past weather, of which only the one-hour-ahead output is used by the agent (Section V-A); forecasting scintillation fading, which would enable adaptive fade reallocation, is future work

Validation: Phase 1 measured $\text{SINR} \in [-11.27, +7.25]$ dB across the full 10–100 km, 24-point angular sweep (worst case at $r = 75$ km, $\theta = 0^\circ$; best case at $r = 10$ km, $\theta = 210^\circ$; Figure 8 and underlying spatial sweep data). Bounds $[-20, +30]$ dB lie 8.7 dB below and 22.8 dB above the observed range, for edge cases (atmospheric fading, relay paths in v2).

2) *Action Space*: Discrete platform choice: $\mathbf{a}_k \in \{0, 1, 2\}$ (HAPS1, HAPS2, HAPS3). UAV relay ($a = 3$) deferred to future work.

3) *Reward Function*: All computations use **linear SINR** ($\gamma_k = 10^{\text{SINR}_{k,\text{dB}}/10}$), never dB arithmetic:

$$R(\mathbf{a}) = 0.5L_{\text{agg}} + 0.3J_{\text{fair}} - 0.2P_{\text{eng}} + \text{Penalties} \quad (43)$$

$$L_{\text{agg}} = \sum_{k=1}^N \log_2(1 + \gamma_k) \quad [\text{bits/s/Hz}] \quad (44)$$

$$J_{\text{fair}} = \frac{(\sum_k \gamma_k)^2}{N \sum_k \gamma_k^2} \in [0, 1] \quad (45)$$

$$P_{\text{eng}} = \frac{1}{3} \sum_{p=1}^3 \left(1 - \frac{\text{SoC}_p}{100} \right) \quad (46)$$

Penalties (thresholds calibrated to Phase 1 measured ranges; aggregated as network average, not per-user):

- Severe outage: $-100 \times$ (fraction of users in outage) if $\text{SINR}_{k,\text{dB}} < -2$ dB (well below average +2 dB)
- Degraded service: $-20 \times$ (fraction of users degraded) if $-2 \leq \text{SINR}_{k,\text{dB}} < 0$ dB (below average)
- Low-battery: -50 if $\text{SoC}_p < 10\%$ with > 0 assigned users

Threshold Rationale: Phase 1 measured: best +7.3 dB, average +2.0 dB, worst -4.3 dB. Penalties trigger below average $\text{SINR}_{k,\text{dB}} < 0$ dB, not within normal operating range (+2 to +7 dB), allowing agent to reward good performance ($> +2$ dB) without false penalties.

D. Critical Implementation Fixes

DQN training must avoid six silent errors:

Error 1: Jain's index on dB (wrong—log scale)

$$J_{\text{wrong}} = \frac{(\sum_k \text{SINR}_{k,\text{dB}})^2}{N \sum_k \text{SINR}_{k,\text{dB}}^2} \quad (47)$$

Fix: Use linear scale

$$J_{\text{correct}} = \frac{(\sum_k 10^{\text{SINR}_{k,10}/10})^2}{N \sum_k (10^{\text{SINR}_{k,10}/10})^2} \quad (48)$$

Error 2: SINR bounds $[0, 100]$ hide outages. **Fix**: Use $\text{SINR} \in [-20, +30]$ dB.

Error 3: HAPS solar from ground rain (HAPS at 20 km is above clouds). **Fix**: Use diurnal cycle only, not weather.

Error 4: LSTM $(24, 2) \rightarrow (60,)$ resolution mismatch. **Fix**: Match resolution: $(24, 2) \rightarrow (1, 2)$ (hourly to hourly).

Error 5: Reward broadcast to all N users identically (the fully-aggregated network reward $R(\mathbf{a})$ given to every user's transition, regardless of that user's own action). With $N = 50$ simultaneously-acting users, any single user's choice moves $R(\mathbf{a})$ by only a small fraction of its value, so the DQN cannot tell which action caused the change — training collapsed to a constant always-pick-HAPS-1 policy (identical evaluation reward/fairness/outage across 200+ consecutive training episodes; verified by inspecting the learned policy's action distribution directly). **Fix**: Decompose the additive components of $R(\mathbf{a}) - L_{\text{agg}}$ and the outage penalties are literally sums (resp. fractions) over users — into a per-user share, so each user is credited with its own throughput/outage contribution. J_{fair} and P_{eng} are genuinely network-level and remain a shared

term. This does not change the total reward mass, only its attribution for learning.

Error 6: HAPS battery SoC update treats a dimensionless energy fraction as if it were already a percentage: $\text{SoC} += (\Delta E_{\text{Wh}}/C_{\text{battery}})$ instead of $\text{SoC} += 100 \times (\Delta E_{\text{Wh}}/C_{\text{battery}})$. This under-scales every SoC update by $100\times$, so battery state barely moves regardless of policy (observed: SoC pinned at $\approx 65\%$ across all three DQN and baseline policies before the fix) – silently disabling the entire energy/battery term of the reward. **Fix:** apply the missing $\times 100$ conversion, and use $C_{\text{battery}} = 12,000$ Wh matching Table I’s value (the module’s own uncorrected default, 1000 Wh, is $12\times$ too small and produces implausible full-charge/full-drain swings within a single hour).

E. HAPS Battery Model

1) *Energy Balance:* State-of-charge updated hourly:

$$\Delta \text{SoC} = \frac{P_{\text{solar}}(t) - (P_{\text{backhaul}} + P_{\text{payload}} + \sum_k P_{k,\text{user}})}{C_{\text{battery}}} \quad (49)$$

where: $P_{\text{backhaul}} = 500$ W (backhaul-link transmitter, HAPS-to-Gateway), $P_{\text{payload}} = 200$ W (control), $P_{k,\text{user}}$ (assigned load), $C_{\text{battery}} = 12000$ Wh.

2) *Solar Power (Diurnal Cycle):* **Critical principle:** HAPS at 20 km (stratosphere, above $> 99\%$ clouds). Solar independent of ground weather.

Solar irradiance (cosine profile):

$$I_0(t_{\text{hr}}) = \begin{cases} 600 \cos(\pi(t_{\text{hr}} - 12)/12) & 6 \leq t_{\text{hr}} \leq 18 \\ 0 & \text{night} \end{cases} \quad [\text{W/m}^2] \quad (50)$$

Generated power:

$$P_{\text{solar}}(t) = 20 \text{ m}^2 \times 0.20 \times I_0(t) = 2400 \text{ W (noon), } 0 \text{ W (night)} \quad (51)$$

3) *Operating Thresholds:* SoC clipped to $[0, 100]\%$:

- **Charging:** Day (6 AM–6 PM) when $P_{\text{solar}} > P_{\text{load}}$
- **Draining:** Night when $P_{\text{solar}} = 0$
- **Critical:** $\text{SoC} < 5\% \Rightarrow$ shutdown (no service)
- **Penalty:** $\text{SoC} < 10\%$ with users $\Rightarrow$ reward -50 (load balancing)

Thresholds: $\text{SoC} \in [0, 100]\%$. Critical: $< 5\%$ (shutdown); penalty: $< 10\%$ (overload). Agent learns to balance users across platforms to avoid over-draining during night.

VI. SUMMARY

A. Channel Model and Interference Characterisation

a) *Propagation metrics and path-loss characterisation.:*

The four core propagation metrics—HAPS free-space path loss, UAV LoS probability, average air-to-ground path loss, and single-node SINR—are implemented and characterised through Figures 2–4.

b) *Multi-HAPS interference and landscape effects.:* The multi-HAPS interference analysis (Figure 8) is landscape-independent for both the 2 GHz service link and the 38 GHz backhaul link: both tiers use pure free-space path loss with no environment-specific term, so suburban, urban, dense-urban, and high-rise environments produce numerically identical SINR (§IV-D). Landscape instead matters for the separate, single-link 2 GHz baseline path-loss analysis (Figure 4): suburban areas retain line-of-sight and sustain usable SINR to greater distances, while high-rise environments show sharp LoS/NLoS transitions and higher blockage loss.

Independent of landscape, adding multi-HAPS co-channel interference degrades the 2 GHz service link by roughly 11–17 dB relative to the noise-only case (comparing the noise-only SNR to the interference-degraded SINR, e.g. $18.5 \rightarrow 7.3$ dB near nadir and $9.9 \rightarrow -7.4$ dB at 50 km; Table A.2, Figure 8).

Backhaul link (38 GHz) does NOT experience this penalty:

The three interfering HAPS are ≈ 65 km from the ground Gateway, placing them far outside the 10–500 m backhaul coverage. Consequently, interference from constellation members remains well below thermal noise floor; backhaul SINR is primarily noise-limited and approximately constant at $+38$ dB, once the HAPS/Gateway directional antenna gains (§II-B) are included in the link budget (Eq. (24), Figure 10).

c) *Scintillation fading (ITU P.618 implementation).:* Figure 3 shows the ITU-R P.618 scintillation fade for the 38 GHz gateway link and the 2.1 GHz UE link. Over $p = 0.01$ – 10% and 10° – 90° elevation, the fade is 0.04–1.8 dB at 38 GHz and 0.007–0.3 dB at 2.1 GHz. The 38 GHz fade is about 0.2 dB at zenith and 1.8 dB at 10° for $p = 0.01\%$. These values replace the earlier empirical fit and its S-band (2.6–4.7 dB) and Ka-band (7.6–13.1 dB) fade values, which were not reproducible from P.618.

Key Findings:

- **9.94 dB variation @ $r = 50$ km:** across the full 24-point angular sweep, SINR ranges -7.40 dB (worst, $\theta = 0^\circ$, toward I1) to $+2.54$ dB (best, $\theta = 210^\circ$)—vastly exceeds distance-based degradation. Note the asymmetric equilateral geometry (I1 @ 0° , I2 @ 60°) means the true best/worst angles are not the cardinal directions shown in Tables A.3a–d; see Figure 7 for the good/bad sector layout.
- **Spatial heterogeneity dominates $r > 30$ km:** At $r = 50$ km, 9.94 dB spatial variation exceeds 2–3 dB per distance doubling.
- **DRL scheduling necessity:** Distance-only assignment fails. Must learn angle-dependent interference patterns.
- **Sector structure:** Good sector (~ 120 – 290° , $\text{SINR} > 0$ dB); bad sector (~ 330 – 75° , $\text{SINR} < -5$ dB within roughly 345 – 60°).

B. DQN Results

The DQN is trained on 400 rolling 24-hour episodes drawn from 2024 and evaluated on 50 calendar days from 2025, which are never seen during training. Each training episode carries the real hourly rain, visibility, and time of day, the

TABLE VII: Policy comparison on the 50 held-out 2025 days. All values are means over five seeds (standard deviation across seeds in brackets).

Policy	Reward per step	Jain	Outage /50
Random (real weather)	-49.1 (1.7)	0.080 (0.008)	20.2 (0.4)
Max-SINR (real weather)	-14.5 (2.1)	0.696 (0.003)	0.9 (0.2)
Random (PreLSTM, synthetic weather)	-49.1 (1.7)	0.080 (0.008)	20.2 (0.4)
Max-SINR (PreLSTM, synthetic weather)	-14.4 (2.1)	0.696 (0.003)	0.9 (0.2)
LSTM + DQN (real weather)	-5.1 (1.1)	0.098 (0.036)	0.0 (0.0)

TABLE VIII: LSTM + DQN across five seeds: mean, standard deviation, median, range, and 95% confidence interval of the mean.

Metric	Mean $\pm$ std	Median	Range	95% CI
Reward per step	-5.11 $\pm$ 1.05	-5.06	[-6.64, -3.67]	[-6.41, -3.80]
Jain fairness	0.098 $\pm$ 0.036	0.084	[0.062, 0.146]	[0.054, 0.143]
Outage /50	0.0 $\pm$ 0.0	0.0	[0, 0]	[0, 0]
Battery SoC (%)	59.95 $\pm$ 0.58	59.89	[59.09, 60.52]	[59.24, 60.67]

ITU-R P.618 rain loss, and the LSTM one-hour-ahead forecast (Section V-A). Each baseline is evaluated on the same 50 days for each seed, and the random and max-SINR baselines use the same evaluation seeds as the DQN. Results are reported over five training seeds: Table VII compares all policies, Table VIII gives the seed-level statistics, Table IX gives the paired comparison with the strongest baseline, and Figure 12 shows the comparison graphically.

a) *Reward and outage.*: LSTM + DQN has the highest mean reward and no outage. Its mean reward is -5.1 against -14.5 for max-SINR, and its outage is zero against 0.9 users on average. Both differences hold on every seed (Table IX).

b) *Fairness.*: Max-SINR is fairer on every seed. Its Jain index is 0.70 against 0.10 for LSTM + DQN. Under the current reward weights (0.5 for throughput, 0.3 for fairness), the learned policy concentrates service on fewer users; a fairness-aware reward is future work.

c) *Baselines and weather.*: Random association is worst on every metric. The real-weather and PreLSTM baselines agree to within 0.1 in reward, because the rain loss enters the reward only through the throughput term, and that term is small for the direct links.

d) *Limitation.*: The DQN chooses among the three HAPS only; UAV relay routing is not in its action space. The topology has a single ground gateway serving the network, and sharing of that gateway is not modelled. The fairness results should be read with this in mind.

VII. CONCLUSION

This document characterised the HAPS-UAV-Gateway air-to-ground channel across three link types – the 2 GHz HAPS→UE service link, the 38 GHz HAPS→Gateway backhaul link, and the two-hop HAPS→UAV→UE relay link

TABLE IX: Paired difference, LSTM + DQN minus Phase 1 max-SINR, per seed and metric. “Seeds” counts the seeds in which the DQN is better.

Metric	Mean difference	95% CI	p-value	Seeds
Reward per step	+9.36	[+7.80, +10.91]	0.0001	5/5
Outage /50	-0.87	[-1.13, -0.61]	0.0007	5/5
Jain fairness	-0.60	[-0.64, -0.56]	< 0.0001	0/5
Battery SoC (%)	+0.02	[-0.05, +0.09]	0.43	4/5

(whose first, 38 GHz hop is the feeder link) – and used that channel model to both characterise multi-HAPS interference (§VI-A) and train a DQN agent for user-platform association (§VI-B).

a) *Channel model.*: Spatial SINR heterogeneity, not distance alone, dominates service-link quality: a 9.94 dB spread at a fixed 50 km radius (§VI-A) exceeds the degradation from doubling distance. The backhaul link remains noise-limited and largely unaffected by constellation geometry, while scintillation adds a frequency- and elevation-dependent fade on both links (§IV-A2).

b) *Relay path analysis.*: The topology’s actual relay route, HAPS→UAV→UE (§IV-E), was evaluated as an alternative to the direct service link. The access hop (UAV→UE) is $\approx +33$ dB SINR at 500 m because the UAV hovers close to the user it serves. The feeder hop (HAPS→UAV, 38 GHz) has antenna gains $G_h = 13.8$ dBi and $G_u = 30.7$ dBi ([14], [15]); its SINR is +37.6 dB at nadir and +22.6 dB at 100 km, well above the +8 dB design floor (Eq. (33), derived from the 16-QAM/CQI-7 minimum-MCS SINR at 10% BLER [13] plus a 3 dB margin). The direct HAPS→UE link, which shares the 2 GHz band with the other two HAPS, falls from +7.3 dB to -9.1 dB. The decode-and-forward end-to-end capacity is 11.1 bits/s/Hz at nadir and 7.5 bits/s/Hz at 100 km, against the direct link’s 2.7 and 0.2 bits/s/Hz (Table XIII). The relay therefore outperforms the direct link at every distance in this model. DQN association agent (§V-B) therefore restricts its action space to the three direct HAPS choices and does not route through the UAV relay, and is left for future work.

c) *Learned association.*: Neither policy dominates: LSTM-fed DQN has the higher reward and lower outage, while max-SINR is fairer. Max-SINR assumes the SINR to every platform is known at decision time, which a deployed system must measure. The LSTM-fed DQN agent (§VI-B) earns the highest reward and the lowest outage of all policies tested on the held-out 2025 days (mean reward -5.1 vs. -14.5 for max-SINR, outage 0 vs. 0.9 users out of 50 on average), with the reward and outage advantage holding in all five training seeds. Max-SINR remains fairer (Jain index 0.70 vs. 0.10), so the learned policy trades fairness for throughput under the current reward. Two directions for future work follow: (1) adding a fairness-aware reward or constraint to close the fairness gap, and testing on harder (e.g. rain- or mobility-heavy) scenarios, and (2) extending the DQN’s action space to include UAV relay routing alongside direct association – now that the relay

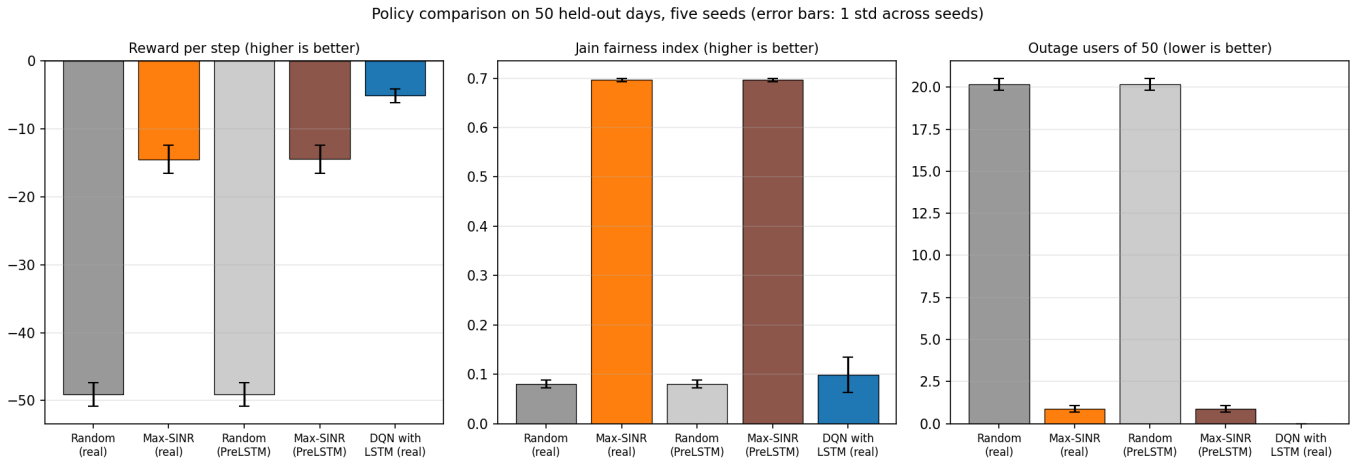


Fig. 12: Policy comparison on the 50 held-out 2025 days over five seeds: reward per step (higher is better), Jain fairness (higher is better), and outage users of 50 (lower is better). Bars are means over seeds; error bars are one standard deviation across seeds.

outperforms the direct link at every distance in this model – so the learned policy, rather than a fixed three-way choice, decides when relaying is worthwhile as a function of user distance.

VIII. ABBREVIATIONS

ABS	Aerial Base Station
AGL	Above Ground Level
CDF	Cumulative Distribution Function
DRL	Deep Reinforcement Learning
DQN	Deep Q-Network
FDMA	Frequency Division Multiple Access
FSPL	Free-Space Path Loss
GW	Gateway
HAPS	High-Altitude Platform Station
ITU	International Telecommunication Union
K-band	18–27 GHz (ITU-R P.618 scintillation model)
Ka-band	27–40 GHz (Scaled from K-band via $(f/20)^{0.7}$)
LoS	Line-of-Sight
LSTM	Long Short-Term Memory
LWC	Liquid Water Content
MDP	Markov Decision Process
MLP	Multi-Layer Perceptron
MVP	Minimum Viable Product
NLoS	Non-Line-of-Sight
PDF	Probability Density Function
PL	Path Loss
RH	Relative Humidity
SINR	Signal-to-Interference-plus-Noise Ratio
SoC	State of Charge (Battery)
TDMA	Time Division Multiple Access
TD	Temporal Difference
UAV	Unmanned Aerial Vehicle
UTD	Uniform Theory of Diffraction

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APPENDIX

A. Core Functions

The listings below show the core functions realising (1)–(13) and the scintillation fade-depth models.

```

1 import numpy as np
2 C = 3e8 # speed of light [m/s]
3
4 # (1) HAPS free-space path loss → eq. (1)
5 def haps_fspl_db(f_mhz, d_km):
6     return 32.44 + 20*np.log10(f_mhz) + 20*np.log10(d_km)
7
8 # (1b) HAPS-to-user/gateway air-to-ground path loss → eq. (1), Arani et al. Eq. 2
9 # Pure free-space path loss: NO landscape/LoS-NLoS term on the HAPS tier
10 # (service link 2 GHz AND feeder link 38 GHz both use this -- see S:IV.C).
11 def haps_a2g_pathloss_db(r_horiz, h_haps, f_hz, h_user=1.5):
12     dh = h_haps - h_user
13     d3d = np.sqrt(r_horiz**2 + dh**2)
14     return 20*np.log10(4*np.pi*f_hz*d3d/C)
15
16 # (2) UAV LoS probability: exact building-blockage series → eq. (2)-(3)
17 # UAV/relay tier ONLY (300 m altitude) -- building blockage does not apply
18 # to the HAPS tier above.
19 def los_probability(r, h_uav, h_user=1.5, alpha=0.3, beta=500.0, xi=15.0):
20     dh = h_uav - h_user
21     J = max(int(np.floor((r/1000.0)*np.sqrt(alpha*beta) - 1.0)), -1)
22     P = 1.0
23     for n in range(J + 1):
24         frac = (n + 0.5) / (J + 1)
25         val = h_uav - frac*dh # h_u - frac*(h_u-h_k)
26         P *= 1.0 - np.exp(-(val**2) / (2*xi**2))
27     return P
28
29 # (3) UAV air-to-ground average path loss → eq. (4)-(5)
30 # UAV/relay tier ONLY -- LoS/NLoS blend via Holis & Pechac (2008).
31 def uav_a2g_pathloss_db(r, h_uav, f_hz=2e9, h_user=1.5,
32     eta_los=1.0, eta_nlos=20.0, **env):
33     dh = h_uav - h_user
34     d3d = np.sqrt(r**2 + dh**2)
35     plos = los_probability(r, h_uav, h_user, **env)
36     base = 20*np.log10(4*np.pi*f_hz*d3d/C)
37     return plos*(base+eta_los) + (1-plos)*(base+eta_nlos)
38
39 # (4) Received power and noise-limited SINR → eq. (6)
40 def sinr_db(p_tx_dbm, pl_db, noise_dbm=-100.0):
41     p_rx = 10**((p_tx_dbm - pl_db)/10.0)
42     return 10*np.log10(p_rx / 10**(noise_dbm/10.0))

```

Listing 1: Core channel-model functions

```

1 # Scintillation fade depth (deterministic) → Equation (12) ITU P.618
2 def scintillation_fade_depth_db(p_time_percent, elevation_deg, freq_ghz=20.0):
3     """ITU P.618 fade depth: D = D_ref * (f/f_ref)^0.7
4         * (1/sin(elev)) * sqrt(log(1/p))"""
5     elev = np.clip(elevation_deg, 5.0, 90.0)
6     freq_factor = (freq_ghz / 20.0) ** 0.7
7     elev_rad = np.radians(elev)
8     elev_factor = 1.0 / np.sin(elev_rad)
9     p_frac = p_time_percent / 100.0
10    time_factor = np.sqrt(np.abs(np.log(p_frac + 1e-6)))
11    ref_fade_db = 2.5 # at p=1%, elev=30deg, f=20GHz
12    return ref_fade_db * freq_factor * elev_factor * time_factor
13
14 # Random fading envelope (log-normal or Gamma)
15 def scintillation_fading_envelope(time_samples, p_time_percent, elevation_deg,
16     freq_ghz=20.0, dist_km=50.0,
17     distribution='log_normal'):
18     """Generate fading multipliers (0-1 linear) matching ITU statistics."""
19     fade_depth_db = scintillation_fade_depth_db(p_time_percent, elevation_deg,
20         freq_ghz)
21     fade_depth_linear = 10.0 ** (-fade_depth_db / 20.0)
22     p_in_fade = p_time_percent / 100.0
23     path_variance_scale = np.sqrt(dist_km / 50.0)
24
25     if distribution == 'log_normal':
26         mu = 0.0
27         sigma = np.sqrt(2.0 * np.log(1.0 / fade_depth_linear)) * \
28             path_variance_scale
29         fading_db = np.random.normal(mu, sigma, time_samples)
30         fading_linear = 10.0 ** (fading_db / 20.0)
31     else: # Gamma
32         shape = 2.0 * path_variance_scale

```

```

33     scale = 1.0 / (shape * (1.0 - p_in_fade))
34     fading_linear = np.random.gamma(shape, scale, time_samples)
35
36     return np.clip(fading_linear, fade_depth_linear, 1.0)
37
38 # Total feeder link loss including scintillation
39 def feeder_link_loss_with_scintillation(distance_km, elevation_deg,
40     freq_ghz=20.0, p_time_percent=1.0, include_scint=True):
41     """ $L_{total} = L_{FSPL} + D(p, \theta, f) \rightarrow \text{eq. (14)}$ """
42     fspl_db_val = 20*np.log10(4*np.pi*freq_ghz*1e9*distance_km*1000 / 3e8)
43     if include_scint:
44         fade_depth_db = scintillation_fade_depth_db(p_time_percent,
45             elevation_deg, freq_ghz)
46     return fspl_db_val + fade_depth_db
47 else:
48     return fspl_db_val

```

Listing 2: Scintillation fading model (ITU P.618)

```

1 # Import functions
2 from atmospheric_losses import (
3     rain_attenuation_db, # ITU-R P.838 rain model
4     fog_attenuation_db, # ITU-R P.840 fog model
5     gas_attenuation_db, # ITU-R P.676 gas absorption
6     scintillation_fading_db, # ITU-R P.618 scintillation
7     total_feeder_loss_db, # Combined sum of all effects
8     get_rainfall_rate_for_availability, # P.837 climatology
9     get_standard_atmosphere, # Reference atmosphere
10 )
11
12 # Example: Link budget at 38 GHz, 11.3° elevation
13 losses = total_feeder_loss_db(
14     rain_rate_mmhr=10.0, # Design rain rate [mm/h]
15     liquid_water_g_m3=0.05, # Fog density [g/m^3]
16     pressure_hpa=1013.0, # Barometric pressure
17     temp_c=15.0, # Temperature
18     rh_percent=60.0, # Relative humidity
19     freq_ghz=38.0, # Frequency
20     elevation_deg=45.0, # Elevation angle
21     time_percentage=0.01, # Outage allowance [%]
22     lat=28.5, lon=77.2 # Location (Delhi)
23 )
24
25 print(f"Total atmospheric loss: {losses['total']:.2f} dB")
26 print(f" Rain: {losses['rain']:.2f} dB")
27 print(f" Fog: {losses['fog']:.2f} dB")
28 print(f" Gas: {losses['gas']:.2f} dB")
29 print(f" Scint: {losses['scintillation']:.2f} dB")

```

Listing 3: Atmospheric losses API (atmospheric_losses.py)

```

1 from atmospheric_losses import (
2     get_rainfall_rate_for_availability,
3     get_rainfall_probability,
4 )
5
6 # Get rain rate for 99.99% link availability (0.01% outage)
7 # This is the "1-in-10000" rain event from 30-year climatology
8 rain_99_99pct = get_rainfall_rate_for_availability(
9     lat=28.5, lon=77.2,
10     availability_percent=99.99
11 )
12
13 # Get probability of rain occurrence
14 prob_rain = get_rainfall_probability(lat=28.5, lon=77.2)
15
16 print(f"Design rain rate: {rain_99_99pct:.1f} mm/h (0.01% exceedance)")
17 print(f"Rain probability: {prob_rain:.1f}% of average year")

```

Listing 4: Climate data API

B. Source Model Parameters (Arani et al. Table 1)

Table X reproduces the system-level simulation parameters from [17] (Table 1, p. 1755) – the primary source cited throughout this document’s channel model. It is included here as a reference for which constants in `channel_model.py` are adopted directly, and which are intentionally not (this project uses its own carrier-frequency plan; see §III.A).

TABLE X: System-level simulation parameters from Arani, Hu & Zhu (2023), Table 1, and their adoption status in this project’s channel model.

Parameter	Value (Arani et al.)	Used here?
HAPS altitude	20 km	Yes – matches HAPS_ALT_M
UAV altitude range ($h_{\min}$, $h_{\max}$)	22.5 m, 150 m	No – plots use 300 m
User height	1.5 m	Yes
Carrier freq. (UAV, HAPS)	28 GHz, 2.11 GHz	No – this project uses its own 2 GHz service/relay and ~38 GHz backhaul/feeder plan (§III.C)
Transmit power (HAPS, UAV)	43 dBm, 24 dBm	Yes – P_TX_HAPS_DBM, P_TX_UAV_DBM
Noise power spectral density	−174 dBm/Hz	No – flat NOISE_DBM=−100 dBm used instead (consistent with ~14 MHz channel BW)
α, β, ξ (LoS probability)	0.1, 750, 8	Yes – matches this document’s “suburban” preset exactly (§III.C); Arani et al. simulated
Reference path loss δ	61.4 dB	No (Phase 2) – calibrated to Arani’s 28 GHz; would need recomputing at this project’s 2
Path-loss exponent η (LoS/NLoS)	2, 3	No (Phase 2) – not yet adopted; current $\eta_{\text{los}}/\eta_{\text{nlos}}$ remain flat dB offsets
Shadowing std. dev. χ (LoS/NLoS)	5.8, 8.7 dB	No (Phase 2) – shadow fading not currently modeled (deterministic mean-value path loss)

C. Data Analysis Tables

The following tables present comprehensive multi-distance and multi-frequency path loss and SINR analysis derived from the channel model using linear-domain calculations per Equations (17) and (11).

TABLE XI: Table A.2: Multi-Distance SINR (2 GHz service link, +X axis from nadir, corrected 65 km equilateral 3-HAPS constellation, $P_{\text{TX,HAPS}} = 43$ dBm). Four distances (0–100 km), SINR degradation from +7.3 dB (nadir) to −9.1 dB (100 km edge). Landscape-independent (pure free-space path loss, §III.C). Demonstrates DRL scheduling necessity. Related figure: Figure 8. Linear-domain calculation per Equation (17).

Distance km	Frequency GHz	Frequency Name	Path Loss Blended dB	Link Type
0.0	2.0	2 GHz (Service)	124.5	Service
10.0	2.0	2 GHz (Service)	125.5	Service
20.0	2.0	2 GHz (Service)	127.5	Service
30.0	2.0	2 GHz (Service)	129.6	Service
50.0	2.0	2 GHz (Service)	133.1	Service
75.0	2.0	2 GHz (Service)	136.3	Service
100.0	2.0	2 GHz (Service)	138.6	Service
0.0	38.0	38 GHz (Backhaul)	150.0	Backhaul
0.1	38.0	38 GHz (Backhaul)	150.0	Backhaul
0.1	38.0	38 GHz (Backhaul)	150.0	Backhaul
0.2	38.0	38 GHz (Backhaul)	150.0	Backhaul
0.3	38.0	38 GHz (Backhaul)	150.0	Backhaul
0.5	38.0	38 GHz (Backhaul)	150.0	Backhaul
0.0	38.0	38 GHz (Feeder, HAPS-UAV)	149.8	Feeder
10.0	38.0	38 GHz (Feeder, HAPS-UAV)	150.9	Feeder
20.0	38.0	38 GHz (Feeder, HAPS-UAV)	153.0	Feeder
30.0	38.0	38 GHz (Feeder, HAPS-UAV)	155.1	Feeder
50.0	38.0	38 GHz (Feeder, HAPS-UAV)	158.6	Feeder
75.0	38.0	38 GHz (Feeder, HAPS-UAV)	161.8	Feeder
100.0	38.0	38 GHz (Feeder, HAPS-UAV)	164.2	Feeder

TABLE XII: Table A.1: Path Loss Comparison (2 GHz service link, user at 1.5 m vs. 38 GHz backhaul link, gateway at 50 m AGL). Eight distances (0–150 km) from HAPS nadir. Pure free-space path loss per Eq. (1) – the HAPS-side links have no landscape-dependent excess loss (§III.C; Arani et al. Eq. 2). Frequency offset $20 \log_{10}(38/2) \approx 25.58$ dB is independent of distance (constant at all distances), derived by canceling distance terms in Eq. (1). Plotted version: Figure 2. Full data in appendix_data/ CSV files.

Distance_km	Serving_HAPS_km	Interferer1_km	Interferer2_km	Desired_Signal_dBm	Interference_dBm	SINR_dB	SNR_dB_Noise_O
0	0	65	32.5	-81.5	-89.1	7.3	1
30	0	65	32.5	-86.6	-85.9	-0.8	1
50	0	65	32.5	-90.1	-82.8	-7.4	
100	0	65	32.5	-95.6	-86.8	-9.1	

D. Relay Path Analysis (HAPS→UAV→UE)

Table XIII gives the full numeric backing for the UAV relay path introduced in §IV-E and modeled by `uav_relay_two_hop_capacity_bps_hz()`: a 38 GHz HAPS→UAV first hop (the feeder link; slant path, UAV hovering at 500 m, with ITU-P.618 scintillation) combined decode-and-forward with a 2 GHz, landscape-dependent UAV→UE second hop. Suburban and high-rise are shown (best/worst case for the landscape-dependent second hop); all four landscapes are in `appendix_data/UAV_relay_path_comparison.csv`.

Landscape	Distance km	Capacity Direct bps Hz	Capacity HAPS UAV bps Hz	Capacity UAV UE bps Hz	Capacity UAV Relay bps Hz
suburban	0	2.67	12.48	11.09	11.09
suburban	10	2.40	12.14	11.09	11.09
suburban	20	1.86	11.43	11.09	11.09
suburban	30	1.37	10.70	11.09	10.70
suburban	50	0.75	9.49	11.09	9.49
suburban	75	0.49	8.37	11.09	8.37
suburban	100	0.46	7.52	11.09	7.52
high rise	0	2.67	12.48	11.09	11.09
high rise	10	2.40	12.14	11.09	11.09
high rise	20	1.86	11.43	11.09	11.09
high rise	30	1.37	10.70	11.09	10.70
high rise	50	0.75	9.49	11.09	9.49
high rise	75	0.49	8.37	11.09	8.37
high rise	100	0.46	7.52	11.09	7.52

TABLE XIII: Table A.4: Direct (HAPS→UE) vs. relay (HAPS→UAV→UE) link comparison, suburban and high-rise landscapes (best/worst case for the landscape-dependent second hop; full 4-landscape data in `appendix_data/UAV_relay_path_comparison.csv`). The relay path is decode-and-forward: the 38 GHz HAPS→UAV first hop (feeder-style slant path, UAV at 500 m, with ITU-P.618 scintillation) is landscape-independent, while the 2 GHz UAV→UE second hop uses the landscape-dependent LoS/NLoS blended model (§III.C). $C_{\text{relay}} = \min(C_{\text{haps-uav}}, C_{\text{uav-ue}})$, implemented in `uav_relay_two_hop_capacity_bps_hz()`.

E. Table A.3: Spatial Heterogeneity Analysis (2D SINR Distribution)

The following analysis extends the multi-distance SINR model to include spatial heterogeneity across the triangular constellation geometry. Users at the same distance from the serving HAPS experience dramatically different SINR values depending on their angular position relative to the interferers, violating the distance-only assumption of classical path-loss models. See Figure 7 (§IV-D) for the constellation geometry and resulting good/bad sectors.

TABLE XIV: Table A.3a: SINR at $\theta = 0^\circ$ (Worst-Case: Toward I1)

r (km)	d_S	d_{I1}	d_{I2}	SINR (dB)
10	10.0	55.0	60.6	5.43
30	30.0	35.0	56.3	-0.83
50	50.0	15.0	58.9	-7.40
75	75.0	10.0	70.5	-11.27
100	100.0	35.0	87.9	-9.06

TABLE XV: Table A.3b: SINR at $\theta = 90^\circ$ (Perpendicular)

r (km)	d_S	d_{I1}	d_{I2}	SINR (dB)
10	10.0	65.8	56.6	5.81
30	30.0	71.6	41.8	0.52
50	50.0	82.0	33.1	-3.89
75	75.0	99.2	37.5	-6.17
100	100.0	119.3	54.5	-6.20

TABLE XVI: Table A.3c: SINR at $\theta = 180^\circ$ (Away from Constellation)

r (km)	d_S	d_{I1}	d_{I2}	SINR
10	10.0	75.0	70.5	7.13
30	30.0	95.0	84.1	4.47
50	50.0	115.0	99.9	2.26
75	75.0	140.0	121.3	0.40
100	100.0	165.0	144.0	-0.92

TABLE XVII: Table A.3d: SINR at $\theta = 270^\circ$ (Opposite Perp.)

r (km)	d_S	d_{I1}	d_{I2}	SINR
10	10.0	65.8	73.8	6.80
30	30.0	71.6	92.2	3.70
50	50.0	82.0	111.1	1.33
75	75.0	99.2	135.3	-0.50
100	100.0	119.3	159.6	-1.70